

Data-Driven Inventory Control with Shifting Demand

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We consider an inventory control problem with lost-sales in a shifting demand environment. Over a planning horizon of T periods, demand distributions can change up to $O(\log T)$ times, but the firm does not know the demand distributions before or after each change, the time periods when changes occur, or the number of changes. Therefore, the firm needs to detect changes and learn the demand distributions only from historical sales data. We show that with censored demand, active exploration in the inventory space is needed for a reasonable detecting and learning algorithm. We provide a theoretical lower bound by partitioning all admissible policies into either exploration-heavy or exploitation-heavy, and for both categories we prove that the convergence rate cannot be better than $\Omega(\sqrt{T})$. We then develop a nonparametric learning algorithm for this problem and prove that it achieves a convergence rate that (almost) matches the theoretical lower bound.

Key words: inventory control, shifting demand, nonparametric learning, censored demand, asymptotic optimality

1. Introduction

In practice, many factors can cause changes in customer demand patterns and consumer behavior. Examples are economic conditions, news and social media, weather and season changes, to name just a few. These information is critical for making inventory ordering decisions (for more discussions on inventory control in changing environments see [Chen and Song 2001](#) and [Chen et al. 2017](#)). However, it is also typical that, a decision maker does not know in advance (1) how many times the demand distribution will change, (2) when these changes will occur, and (3) what the demand structure will be before and after each change. These are exactly the settings of this paper.

We study a stochastic inventory control problem with lost-sales over a planning horizon T . At the beginning of every period t , the firm makes an inventory order-up-to decision y_t to satisfy random demand. The demand distribution is unknown and can change up to $O(\log T)$ times throughout the planning horizon. In particular, the demand for type v in period t is D_t^v , and it needs to be satisfied as much as possible by y_t . If supply and demand do not match, there is a holding cost h for any unit of leftover inventory and a penalty cost b for any lost-sales. Lostsales are not observable

and the firm is only able to access censored demand data. The firm's goal is to detect demand changes and learn the demand distributions only from historical sales data, and to minimize the T period total cost by making good decisions of y_t . We develop a data-driven algorithm for this problem and prove that its performance converges to that of the optimal policy of a clairvoyant who had complete information of the demand distributions and changes, and we also provide the convergence rate of the algorithm.

1.1. Contributions and Challenges

We summarize our contributions in the context of the challenges from the following aspects.

1. **Introducing nonstationarity to data-driven inventory control.** Related to this work, there is a vast literature on stochastic inventory control with nonparametric demand learning (e.g., [Levi et al. 2015](#), [Cheung and Simchi-Levi 2018](#), [Zhang et al. 2018](#), [Ban and Rudin 2018](#), [Zhang et al. 2019](#), [Ban 2019](#)), but all of them assume stationary demand over the entire planning horizon. When demand is allowed to change, there are no existing lower bound on the learning rate, nor is there a learning algorithm that attains a matching upper bound. In this paper, we fill these two gaps with two novel results.

- (a) **Proof of the theoretical lower bound of $\Omega(\sqrt{T})$.** To develop the lower bound for convergence rate of any admissible learning algorithm, we introduce two categories of policies, namely, the exploration-heavy policy and the exploitation-heavy policy. These two categories are exhaustive in the sense that any admissible learning policy must fall into either of them. Exploration-heavy policies behave the worst when the demand changes occur toward the end of the planning horizon, because they will keep spending efforts on exploration and change detection all over the planning horizon, resulting in a cost increase of $\Omega(\sqrt{T})$. On the other hand, exploitation-heavy policies spend not enough effort on exploration so that the probability of mistaking one demand distribution for another can be significant. The failure to detect demand changes can cause a cost increase of $\Omega(\sqrt{T})$. Summarizing these two scenarios, the regret of any admissible policy is lower bounded by $\Omega(\sqrt{T})$ no matter which category it belongs to.

Note that when demand is stationary, the lower bound for convergence rate is $\Omega(\log T)$ ([Huh and Rusmevichientong 2009](#), [Besbes and Muharremoglu 2013](#), [Chen and Chao 2019](#)). Comparing the two lower bounds, it shows that the complexity caused by a shifting environment is significant.

- (b) **Development of a data-driven algorithm with convergence rate $O(\sqrt{T}(\log T)(\log \log T)^4)$.** The learning algorithm for inventory control with shifting

demand (LAIS for short) detects demand changes and updates demand estimations following cycles of equal length. When making ordering decisions, LAIS discounts all the sales data from previous cycles, because the demand distribution may have changed since then. Sales data contain downward biases since the demand data is truncated, but despite this, LAIS achieves a regret rate of $O(\sqrt{T}(\log T)(\log \log T)^4)$, matching the theoretical lower bound up to a $\log T$ term. This is the first tight bound for inventory control problems with censored demand in a changing environment.

2. **Learning without observing objective or gradient feedbacks.** In the operations management literature that explores the learning problem in a nonstationary environment (Besbes and Zeevi 2011, Besbes et al. 2015, Keskin and Zeevi 2016, Chen et al. 2019, Cheung et al. 2019, Keskin et al. 2019), all of the studies assume that after taking an action, the decision maker has access to either the objective (zeroth order) feedback or the gradient (first order) feedback, the two most commonly assumed feedbacks in the learning literature. But this assumption is not satisfied in our problem. Due to demand censoring, the decision maker is not able to observe the realized objective value since lost-sales quantity (and thus lost-sale penalty) is not observable. Furthermore, the realized gradient value is not observable either. In order to conduct stochastic gradient decent, the realized right gradient is needed, which is $-b + (b+h)\mathbb{1}(D_t^v \leq y_t)$ (Huh and Rusmevichientong 2009). But this quantity is not observable, because the event $\mathbb{1}(D_t^v = y_t)$ cannot be differentiated from $\mathbb{1}(D_t^v > y_t)$ (under both events we only observe stockouts).

Moreover, all the aforementioned research works are oblivious to the data censoring issue - one of the major challenges in this work.

1.2. Literature Review

Inventory control has been studied in the literature under nonstationary environment with complete information. In one of the settings, demand is modulated by an underlying Markov chain that represents the state of the world. Song and Zipkin (1993) consider a continuous review system and prove that the world-dependent order-up-to policy is optimal without setup cost and the world-dependent (r, S) policy is optimal with setup cost. Cheng and Sethi (1999) then study the discrete review system with setup cost and prove that the world-dependent (s, S) policy is optimal. The serial inventory system with Markov modulated demand is studied in Chen and Song (2001) and more recently in Chen et al. (2017), where the theoretical properties and computation issues of the optimal solution are studied. Treharne and Sox (2002) assume the demand distributions and the Markov transition probabilities are not known and model the problem as a POMDP. They

propose several heuristics to solve the problem, but convergence rate is not analyzed. [Graves \(1999\)](#) consider the demand process to be an integrated moving average process. [Karlin \(1960\)](#), [Zipkin \(1989\)](#) and [Kapusinski and Tayur \(1998\)](#) explore the situation where demand changes periodically over cycles and study the properties of the optimal solution. [Xiao et al. \(2015\)](#) study the joint pricing and inventory control problem under a fluctuating cost structure.

Data-driven inventory control problems have received increasingly more attention in the recent decade. Considering the repeated (single period) newsvendor problem under learning, [Levi et al. \(2007\)](#) and [Levi et al. \(2015\)](#) provide offline learning algorithms using sample average approximation (SAA). [Huh et al. \(2011\)](#) study the online version using Kaplan-Meier estimator and [Besbes and Muharremoglu \(2013\)](#) compare continuous and discrete demands using properties of the newsvendor solution. The offline multiperiod stochastic inventory control problem is studied in [Cheung and Simchi-Levi \(2018\)](#) using SAA, and the online problem is solved by [Huh and Rusmevichientong \(2009\)](#) using stochastic gradient descent. The lost-sale system with positive lead time is considered by [Huh et al. \(2009\)](#) and [Zhang et al. \(2019\)](#), warehouse capacity for multiple products is considered in [Shi et al. \(2016\)](#), inventory control for perishable products is studied in [Zhang et al. \(2018\)](#), systems with fixed setup cost is studied in [Yuan et al. \(2019\)](#), inventory control with stockout substitution is considered in [Chen and Plambeck \(2008\)](#) and [Chen and Chao \(2019\)](#), the big data newsvendor model with feature information is studied in [Ban and Rudin \(2018\)](#), managing new product with covariate information is studied in [Ban et al. \(2018\)](#), learning in the dual sourcing inventory system is considered by [Chen and Shi \(2019\)](#), and confidence intervals for inventory decisions using censored data is studied in [Ban \(2019\)](#). All of the studies here consider stationary environment, i.e., the demand distribution stays the same throughout the planning horizon.

In the operations management literature, there are a number of papers that consider nonstationarity and learning. [Besbes and Zeevi \(2011\)](#) study a dynamic pricing problem where the underlying demand function changes once. [Keskin and Zeevi \(2016\)](#) consider the scenario that the demand function changes throughout the planning horizon subject to a budget of total changes. [Keskin et al. \(2019\)](#) consider the joint pricing and inventory control problem with perishable products in a changing environment. They assume lost-sales are observable, therefore demand is uncensored. When demand is parametric, the regret of their algorithm is $O(T^{1/2}(\log T))$, and when the noise is nonparametric, the regret is $O(T^{2/3}(\log T)^{1/2})$. In more general settings, [Besbes et al. \(2015\)](#) study the problem of nonstationary stochastic optimization with a budget of changes and develop both upper and lower bounds for regret. The setting is further explored by [Chen et al. \(2019\)](#) adopting a more general measure for the change of the demand function. [Cheung et al. \(2019\)](#) consider the bandit setting and relax the assumption that the budget for changes needs to be known by the

decision maker. All of the papers above assume function feedback is available. Besbes et al. (2015) and Chen et al. (2019) also consider the case where gradient feedback is available, and implement the online gradient decent procedure. For our problem, as mentioned previously, neither function feedback nor gradient feedback is observable due to demand censoring.

1.3. Organization

The rest of the paper is organized as follows: Section 2 formally formulates the problem; Section 3 provides the theoretical lower bound; Section 4 introduces the data-driven algorithm; Section 5 presents a numerical study. The paper concludes in Section 6. All technical proofs are relegated to the Appendix.

Throughout the paper, for $x \in \mathbb{R}$, $\lceil x \rceil$ denotes the smallest integer greater than or equal to x , and $\lfloor x \rfloor$ denotes the largest integer smaller than or equal to x .

2. Problem Formulation

We consider an inventory control problem over T periods. Demand in each period is a discrete random variable. Different from the literature that usually assumes demand to be independent and identically distributed (i.i.d.), we allow the demand distribution to change over time.

Demand process. Suppose the demand distribution changes V times during the planning horizon $\{1, \dots, T\}$, where V is unknown and upper bounded by $\beta \log T$ for some constant $\beta > 0$. For $v = 0, \dots, V$, let D_t^v denote the v th type of demand at period t , whose probability mass function is $\mathbf{w}^v = (w^{v0}, w^{v1}, \dots)$ with

$$\mathbb{P}(D_t^v = m) = w^{vm}, \quad \text{for } m = 0, 1, \dots$$

Assume $\mathbb{E}[D_t^v] > 0$ and $\text{Var}[D_t^v] < +\infty$. Since no parametric form is assumed, the model for the distribution of D_t^v is nonparametric. The demand process starts with $\mathbf{w}^0$, and let $1 < l^1 < \dots < l^V \leq T$ be the time periods when the 1st, $\dots$ and V th changes occur. Denote $l^0 = 1$ and $l^{V+1} = T + 1$. At the beginning of period l^v , demand distribution changes from $\mathbf{w}^{v-1}$ to $\mathbf{w}^v$, and demands during periods $l^v, \dots, l^{v+1} - 1$ are i.i.d.. We denote

$$\bar{\mathbf{w}} = \{\mathbf{w}^0, \dots, \mathbf{w}^V\}, \quad \bar{\mathbf{l}} = \{l^1, \dots, l^V\}.$$

The firm does not know the number of changes V , the periods of the changes $\bar{\mathbf{l}}$, or the values of $\bar{\mathbf{w}}$. For $v = 0, \dots, V$, we use the generic form D^v and D_t^v interchangeably when no confusions may arise.

Let $\mathcal{Y} = \{0, 1, \dots, \tilde{y}\}$ be the set of order-up-to levels, and x_t and $y_t \in \mathcal{Y}$ denote the inventory levels before and after ordering at the beginning of period t respectively. Demands are satisfied as much as possible by on-hand inventory, and unsatisfied demands are lost and not observable to the firm. During period t , the firm observes the sales quantity $s_t = \min\{D_t^v, y_t\}$, which is censored demand.

System dynamics and objective function. We assume that the system is initially empty, i.e., $x_1 = 0$. Let $\mathcal{H}_t$, $t \geq 0$, denote the filtration generated by sales and decisions made up to time t , i.e., $\mathcal{H}_t = \sigma(y_r, s_r : r = 1, \dots, t)$ and $\mathcal{H}_0 = \emptyset$. An admissible policy is represented by a sequence of order-up-to levels, $\{y_t(\mathcal{H}_{t-1}), t = 1, \dots, T\}$ with $y_t(\mathcal{H}_{t-1}) \in \mathcal{Y}$ and $y_t(\mathcal{H}_{t-1}) \geq x_t$, where y_t is a mapping from the space of $\mathcal{H}_{t-1}$ to $\mathcal{Y}$, that is, y_t depends only on $\mathcal{H}_{t-1}$ under censored demand. Sometimes we make the dependence on $\mathcal{H}_{t-1}$ implicit when no confusions may occur.

Given an admissible policy π , we describe the sequence of events for each period $t \in \{l^v, \dots, l^{v+1} - 1\}$, $v = 0, \dots, V$, as follows. Note that x_t and y_t are dependent on policy π , though our notation makes the dependence implicit.

- (a) At the beginning of period t , the firm observes the starting inventory levels x_t .
- (b) The firm determines an ordering quantity to bring the inventory levels to y_t . The unit ordering cost for each product is normalized to 0 without loss of generality. Order lead times are zero, i.e., any order placed is received and can be used to satisfy demand in the same period.
- (c) Demand realizes and is satisfied by on-hand inventory as much as possible.
- (d) At the end of period t , the following cost incurs

$$h(y_t - D_t^v)^+ + b(D_t^v - y_t)^+,$$

where h is the holding cost for one unit of leftover inventory and b is the shortage cost every time a demand is not satisfied.

The firm aims to construct an admissible policy $\{y_t, t = 1, \dots, T\}$ with $y_t \geq x_t$ and $x_t = (y_{t-1} - D_{t-1}^v)^+$, such that the expected T -period total profit in (1) is maximized:

$$\sum_{v=0}^V \sum_{t=l^v}^{l^{v+1}-1} (h\mathbb{E}[(y_t - D_t^v)^+] + b[(D_t^v - y_t)^+]). \quad (1)$$

Since the underlying demand is not known, the firm cannot solve the problem in (1) directly, necessitating the development of a learning algorithm that detects the changes and updates the demand and substitution estimations on the fly to facilitate inventory decisions.

Clairvoyant optimal solution. If the firm knew the complete information of the problem *a priori*, i.e., it knew the values of V , $\bar{\mathbf{I}}$ and $\bar{\mathbf{w}}$, then it could solve the problem in (1) directly. Denote the clairvoyant optimal policy by π^* , and the optimal expected T -period profit as $\mathcal{Q}^{\pi^*}(V, \bar{\mathbf{I}}, \bar{\mathbf{w}}, T)$.

We first consider a problem with stationary environment. Suppose that, from period l^v to $l^{v+1} - 1$, $0 \leq v \leq V$, the demand distribution $\mathbf{w}^v$ is known. Then the clairvoyant single period optimization problem for this interval of periods is

$$\min_{y \in \mathcal{Y}} Q(\mathbf{w}^v, y) = h\mathbb{E}[(y - D^v)^+] + b\mathbb{E}[(D^v - y)^+]. \quad (2)$$

Let the clairvoyant optimal solution of (2) be y^{v*} , and if there are multiple optimizers, let y^{v*} be the smallest one. If the firm is able to reach y^{v*} for every period from l^v to $l^{v+1} - 1$, it would attain the optimal expected profit in every period, and the expected T -period total profit would be

$$\tilde{\mathcal{Q}}(V, \bar{\mathbf{I}}, \bar{\mathbf{w}}, T) = \sum_{v=0}^V \sum_{t=l^v}^{l^{v+1}-1} Q(\mathbf{w}^v, y^{v*}).$$

But because of carry-over inventories, $y_t = \max\{x_t, y^{v*}\}$, which implies that y^{v*} may not be achieved for some periods from l^v to $l^{v+1} - 1$. Therefore, $\tilde{\mathcal{Q}}(V, \bar{\mathbf{I}}, \bar{\mathbf{w}}, T)$ may not be achievable, yielding

$$\tilde{\mathcal{Q}}(V, \bar{\mathbf{I}}, \bar{\mathbf{w}}, T) \leq \mathcal{Q}^{\pi^*}(V, \bar{\mathbf{I}}, \bar{\mathbf{w}}, T).$$

Performance metric. Without complete information about demand, for an admissible data-driven policy π , its expected profit is

$$\mathcal{Q}^{\pi}(V, \bar{\mathbf{I}}, \bar{\mathbf{w}}, T) = \sum_{v=0}^V \sum_{t=l^v}^{l^{v+1}-1} \mathbb{E}[Q(\mathbf{w}^v, y_t)].$$

Define the regret of π as

$$\begin{aligned} \mathcal{R}^{\pi}(\Lambda, T) &= \max_{V \in \{0, \dots, \lfloor \beta \log T \rfloor\}} \max_{1 < l^1 < \dots < l^V \leq T} \max_{\substack{\mathbf{w}^v \in \Lambda \\ v=0, \dots, V}} \{ \mathcal{Q}^{\pi}(V, \bar{\mathbf{I}}, \bar{\mathbf{w}}, T) - \mathcal{Q}^{\pi^*}(V, \bar{\mathbf{I}}, \bar{\mathbf{w}}, T) \} \\ &\leq \max_{V \in \{0, \dots, \lfloor \beta \log T \rfloor\}} \max_{1 < l^1 < \dots < l^V \leq T} \max_{\substack{\mathbf{w}^v \in \Lambda \\ v=0, \dots, V}} \{ \mathcal{Q}^{\pi}(V, \bar{\mathbf{I}}, \bar{\mathbf{w}}, T) - \tilde{\mathcal{Q}}(V, \bar{\mathbf{I}}, \bar{\mathbf{w}}, T) \}, \end{aligned}$$

where Λ is the set of potential demand distributions that satisfy Assumption 1, which will be introduced shortly. As it can be seen, regret measures the cost increase due to lack of demand information, and a smaller value of regret means a more cost-efficient policy π .

ASSUMPTION 1. (1) For any demand distribution $\mathbf{w}^v \in \Lambda$, there exist some integer $\tilde{m} \geq 0$ and constants $\theta_1 > 0$, $\theta_2 > 0$, such that, $\mathbb{P}(D^v = m) \leq \theta_1 \exp(-\theta_2 m)$ for any $m \geq \tilde{m}$.

(2) For any $\mathbf{w}^v \in \Lambda$ and suboptimal $y \in \mathcal{Y}$ under $\mathbf{w}^v$, there exists a constant $\delta > 0$ such that $Q(\mathbf{w}^v, y^{v*}) \leq Q(\mathbf{w}^v, y) - \delta$.

Assumption (1) requires the demand distributions to be light-tailed, which is satisfied by common discrete distributions such as Poisson, Geometric, Binomial, and all other discrete random variables with bounded support. Assumption (2) requires that for any suboptimal $y \in \mathcal{Y}$, its cost is a constant away from the true optimal cost.

3. Lower Bound

In this section, we will first present the theoretical lower bound for $\mathcal{R}^\pi(\Lambda, T)$, and then explain the roadmap of the proof.

THEOREM 1. *Under Assumption 1, for any admissible policy π and $T \geq 3$, one has*

$$\mathcal{R}^\pi(\Lambda, T) \geq \frac{1}{40e} \sqrt{T}.$$

This theorem demonstrates that the lower bound of regret for inventory control under censored demand in a shifting environment is $\Omega(\sqrt{T})$. This is very different from the lower bound in a stationary environment, which is shown to be $\Omega(\log T)$ (Huh and Rusmevichientong 2009, Besbes and Muharremoglu 2013, Chen and Chao 2019). Therefore, detection of demand changes is indeed a critical task when developing learning policies.

Theorem 1 is proved based on the idea that if an algorithm spends too much effort on exploring and detecting the changes, it will sacrifice time for collecting profits. On the other hand, if it spends too little effort on detecting the changes, there will always be a risk of implementing the wrong solutions for too long without noticing the environment changes. In the proof we first show that there exists some problem instance that requires active exploration to detect demand changes (Section 3.1). Then we show in Section 3.2 that for this specific problem instance, any admissible learning algorithm, even if it optimally balances the exploration-exploitation trade-off, the algorithm cannot achieve a regret lower than $\Omega(\sqrt{T})$. In order to quantify the extent of exploration and exploitation, we introduce exploration-heavy policies and exploitation-heavy policies.

3.1. Exploration is Needed due to Demand Censoring

In this section, we present a specific problem instance to show that, with censored demand, it indeed requires exploration in the inventory space to detect the demand change and learn the demand

distribution. Note that in the dynamic pricing literature, studies have shown the necessity of active exploration to learn the demand-price function for uncensored demand (den Boer and Zwart 2013, Keskin and Zeevi 2014), and here we echo this insight using an application in stochastic inventory control with censored demand.

The specific example is as follows. Let the support for demand be $\{0, 1, 2\}$, and the set of potential demand distributions be $\Lambda = \{\mathbf{w}^0 = (3/5, 1/5, 1/5), \mathbf{w}^1 = (1/5, 1/5, 3/5)\}$. The number of demand changes $V = 1$, and the demand distribution switches from $\mathbf{w}^0$ to $\mathbf{w}^1$ at the beginning of period l^1 , $1 < l^1 \leq T$. Cost parameters are $b = 1, h = 1$, and inventory decisions are $y_t \in \mathcal{Y} = \{0, 1, 2\}$. Next we further reduce the complexity of the problem in two regards. First, assume the firm knows that the demand changes from $\mathbf{w}^0$ to $\mathbf{w}^1$ but does not know the time of change l^1 . Second, we remove the carry-over inventory constraint $y_t \geq (x_{t-1} - D_{t-1}^v)^+$, so that the problem is relaxed from the original constrained learning problem to an unconstrained problem. Any admissible policy for the original problem is still admissible for this relaxed problem, therefore, a lower bound of regret for the latter must also be a lower bound for the original problem which carries less information and more constraints.

For $v = 0, 1$, consider the following single period problem,

$$\min_{y \in \mathcal{Y}} Q(\mathbf{w}^v, y) = \min_{y \in \mathcal{Y}} \{\mathbb{E}[D^v - y]^+ + \mathbb{E}[y - D^v]^+\}.$$

First of all,

$$Q(\mathbf{w}^0, 0) = \frac{3}{5}, \quad Q(\mathbf{w}^0, 1) = \frac{4}{5}, \quad Q(\mathbf{w}^0, 2) = \frac{7}{5}, \quad (3)$$

therefore under $\mathbf{w}^0$, the optimal solution $y^{0*} = 0$. On the other hand,

$$Q(\mathbf{w}^1, 0) = \frac{7}{5}, \quad Q(\mathbf{w}^1, 1) = \frac{4}{5}, \quad Q(\mathbf{w}^1, 2) = \frac{3}{5}, \quad (4)$$

and under $\mathbf{w}^1$, the optimal solution $y^{1*} = 2$. The clairvoyant solution is to implement $y_t = 0$ for periods 1 to $l^1 - 1$, and to implement $y_t = 2$ for periods l^1 to T . But l^1 is unknown, and the firm needs to detect it by observing sales data.

Because demand is censored, the sales data depends on the inventory level. Under $\mathbf{w}^v$, $v = 0, 1$, when inventory order-up-to level is y_t , the distribution for sales quantity s_t is the following: for $y_t = 0$,

$$\mathbb{P}(s_t = 0 | y_t = 0) = 1, \quad (5)$$

which does not depend on the demand type v ; for $y_t \geq 1$,

$$\mathbb{P}(s_t | y_t \geq 1) = \begin{cases} w^{v s_t} & \text{if } s_t \leq y_t - 1, \\ 1 - \sum_{m=0}^{y_t-1} w^{v m} & \text{if } s_t = y_t. \end{cases} \quad (6)$$

From (5) one can see that the inventory level $y_t = 0$ is an uninformative inventory level, because sales quantity will always be 0 under both $\mathbf{w}^0$ and $\mathbf{w}^1$ if setting $y_t = 0$. In other words, if the firm keeps implementing $y_t = 0$, which is the optimal solution for $\mathbf{w}^0$, it will not be able to detect the shifting point of the demand distribution. In (6), one observes that when $y_t \geq 1$, the sales distribution depends on v , which means that if the firm deviates from y^{0*} , it will potentially be able to differentiate $\mathbf{w}^1$ from $\mathbf{w}^0$ and to detect the shifting point. This example shows that active exploration is needed to detect the shifting point, a phenomenon that will not occur if demand is uncensored.

REMARK 1. Note that $y_t = 0$ is not the only value that can be an uninformative inventory level, and we just propose this specific instance for the ease of presentation. Another instance would be $\Lambda = \{\mathbf{w}^0 = (1/5, 2/5, 2/5), \mathbf{w}^1 = (1/5, 1/5, 3/5)\}$, in which case $y^{0*} = 1$ and $y^{1*} = 2$. Here, both $y_t = 0$ and $y_t = 1$ are uninformative inventory levels, because by implementing either one of them, the distribution for sales quantity is the same regardless of the underlying demand distribution being either type 0 or 1. Therefore, the firm still needs to deviate from $y^{0*} = 1$ to detect the demand change.

3.2. Exploration-Heavy Policies and Exploitation-Heavy Policies

In this section we introduce exploration-heavy policies and exploitation-heavy policies for the specific problem instance discussed in Section 3.1, for which exploration is needed. The question then comes to how often and by how much the firm should deviate from y^{0*} and explore other inventory levels. We partition all admissible policies into two categories: exploration-heavy policies and exploitation-heavy policies. As the names suggest, exploration-heavy policies spend more time and effort on exploration and deviation from y^{0*} than the exploitation-heavy ones. We then show that no matter which category an admissible policy belongs to, the regret for it is lower bounded by $\Omega(\sqrt{T})$. Next we introduce the definitions for the two categories of policies.

Divide the planning horizon $\{1, 2, \dots, T\}$ into batches of $\lceil \sqrt{T} \rceil$ periods, so there are in total $N = \lceil T / \lceil \sqrt{T} \rceil \rceil$ batches. Let $\mathbb{P}_j^\pi$ be the distribution for sales quantities $s_1, \dots, s_T$ under policy π when the demand change occurs at the beginning of batch j , based on which the expectation $\mathbb{E}_j^\pi$ is defined.

Let $\mathcal{K}(\mathbb{P}_{j+1}^\pi, \mathbb{P}_j^\pi)$ denote the Kullback-Leibler (KL) divergence between $\mathbb{P}_{j+1}^\pi$ and $\mathbb{P}_j^\pi$, which is defined by

$$\mathcal{K}(\mathbb{P}_{j+1}^\pi, \mathbb{P}_j^\pi) = \mathbb{E}_{j+1}^\pi \left[\log \frac{\mathbb{P}_{j+1}^\pi(s_t : 1 \leq t \leq T)}{\mathbb{P}_j^\pi(s_t : 1 \leq t \leq T)} \right].$$

An admissible policy π is said to be exploration-heavy, if $\mathcal{K}(\mathbb{P}_{j+1}^\pi, \mathbb{P}_j^\pi) > 1$ for all $j = 1, \dots, N-1$. Otherwise, π is exploitation-heavy. One extreme case of the exploitation-heavy policy is implementing $y^{0*} = 0$ at every period $t = 1, \dots, T$, which leads to $\mathcal{K}(\mathbb{P}_{j+1}^\pi, \mathbb{P}_j^\pi) = 0$ for all $j = 1, \dots, N-1$. We provide regret lower bound for both categories of policies in the following two propositions.

PROPOSITION 1. *For any exploration-heavy policy π and $T \geq 3$, one has*

$$\mathcal{R}^\pi(\Lambda, T) \geq \sqrt{T} - 1.$$

The intuition to prove Proposition 1 is the following. For an exploration-heavy policy π , the best scenarios are when the demand change happens early of the planning horizon. This is because in such scenarios the learning policy can quickly detect the switch and adjust ordering decisions accordingly, and for the rest of the planning horizon, it does not need to pay much attention to change detection any more. A more challenging situation is when the demand change occurs at the end of the planning horizon, say, at the beginning of batch N . In this case, an exploration-heavy policy needs to keep exploring non-optimal inventory levels for change detection at least up to this point of time. Under $\mathbf{w}^0$, if inventory levels other than $y^{0*} = 0$ are implemented, as seen in (3), the objective cost will increase by at least $1/5$. Proposition 1 suggests that if the demand change occurs toward the end of the planning horizon, any exploration-heavy policy will actively explore at non-optimal inventory levels for at least $\Omega(\sqrt{T})$ periods in expectation.

Let n_j be the first period of batch j , i.e., $n_j = (j-1)\lceil\sqrt{T}\rceil + 1$ for $j = 1, \dots, N$. Technically, the key to proving Proposition 1 is to establish the following inequalities when the demand change happens at the start of batch N .

$$\begin{aligned} \mathcal{R}^\pi(\Lambda, T) &= \mathbb{E}_N^\pi[\mathcal{Q}^\pi(V, \bar{\mathbf{I}}, \bar{\mathbf{w}}, T) - \tilde{\mathcal{Q}}(V, \bar{\mathbf{I}}, \bar{\mathbf{w}}, T)] \\ &\geq \sum_{j=1}^{N-1} \mathbb{E}_N^\pi \left[\sum_{t=n_j}^{n_{j+1}-1} \frac{1}{5} |y^{0*} - y_t(\mathcal{H}_{t-1})| \right] \end{aligned} \quad (7)$$

$$= \sum_{j=1}^{N-1} \mathbb{E}_{j+1}^\pi \left[\sum_{t=n_j}^{n_{j+1}-1} \frac{1}{5} |y^{0*} - y_t(\mathcal{H}_{t-1})| \right] \quad (8)$$

$$\geq \sum_{j=1}^{N-1} \mathcal{K}(\mathbb{P}_{j+1}^\pi, \mathbb{P}_j^\pi). \quad (9)$$

In the above inequalities, (7) is because every time the inventory level deviates from y^{0*} before period n_N there is a cost increase of at least $1/5$ (as shown in (3)). Equality (8) follows from the fact that for periods before n_{j+1} , available historical informations are identical for either the demand changes at period n_{j+1} or n_N . (9) is critical in that it builds the connection between cost increase and the KL divergence. We prove this inequality separately for $y_t(\mathcal{H}_{t-1}) = 0, 1$ and 2, by analyzing the distribution of sales quantity s_t conditional on the value of $y_t(\mathcal{H}_{t-1})$. See the detailed and formal proof in the appendix.

PROPOSITION 2. *For any exploitation-heavy policy π and $T \geq 3$, one has*

$$\mathcal{R}^\pi(\Lambda, T) \geq \frac{1}{40e} \sqrt{T}.$$

An exploitation-heavy policy π spends relatively less effort on exploration in the inventory space, therefore, there exists a higher risk that it is not able to detect the change point successfully. To be more specific, let j_0 be the period such that $\mathcal{K}(\mathbb{P}_{j_0+1}^\pi, \mathbb{P}_{j_0}^\pi) \leq 1$, which means that under an exploitation-heavy policy π , the two distributions for sales quantities $\mathbb{P}_{j_0+1}^\pi$ and $\mathbb{P}_{j_0}^\pi$ do not differ much. Then we show that the probability for policy π to mistake one for the other is at least $1/(4e)$.

Intuitively, if the belief of the change point is one batch away from the true change point, this will potentially lead to mistreatment of at least one batch of periods, which should result in $\Omega(\sqrt{T})$ cost increase. Indeed, we show that

$$\max \left\{ \mathbb{E}_{j_0}^\pi \left[\sum_{t=n_{j_0}}^{n_{j_0+1}-1} |y^{1*} - y_t(\mathcal{H}_{t-1})| \right], \mathbb{E}_{j_0+1}^\pi \left[\sum_{t=n_{j_0}}^{n_{j_0+1}-1} |y^{0*} - y_t(\mathcal{H}_{t-1})| \right] \right\} > \frac{1}{8e} \lceil \sqrt{T} \rceil. \quad (10)$$

Note that y^{0*} is optimal for periods $n_{j_0}, \dots, n_{j_0+1}-1$ when demand change happens at the beginning of batch j_0+1 , and y^{1*} is optimal for these periods when demand changes at the beginning of batch j_0 . Therefore, (10) shows that, for an exploitation-heavy policy π , the expected deviation from the optimal solution is lower bounded by $1/(8e) \lceil \sqrt{T} \rceil$. By (3) and (4), once deviation happens, it will result in at least a cost of $1/5$. Thus, policy π will incur a regret of at least $1/(40e) \sqrt{T}$. A detailed and formal proof is provided in the appendix.

Combing Propositions 1 and 2, Theorem 1 then follows.

4. Data-Driven Algorithm and Matching Upper Bound

In this section, we first develop a learning algorithm that detects the changes of demand along the way and updates their estimations accordingly. Notice that the algorithm is designed for this

general class of problems, not just for the specific example presented in Section 3. Then we provide upper bounds for the estimation error and prove the convergence rate of the learning algorithm.

4.1. Learning Algorithm for Inventory Control with Shifting Demand

We refer to the algorithm as learning algorithm for inventory control with shifting demand (LAIS for short). The idea of LAIS is suggested by the proof of lower bound in Theorem 1. To be more specific, LAIS proceeds in batches, with each batch containing $O(\sqrt{T})$ periods. During batch i , LAIS spends a small fraction of periods on active exploration in the inventory space and updates its estimator $\hat{\mathbf{w}}_i$. Then $\hat{\mathbf{w}}_i$ is used for decision making in the rest of the batch.

Algorithm 1 Learning Algorithm for Inventory Control with Shifting Demand (LAIS)

The LAIS algorithm runs in batches, with each batch containing L exploration and detection periods, followed by I exploitation periods. We set

$$I = \lceil \sqrt{T} \rceil, \quad L = \lceil \log I (\log \log I)^3 \rceil, \quad t_i = (i-1)(L+I),$$

where $t_i + 1$ is the first period of batch i . There are a total of $M = \mathcal{O}(\sqrt{T})$ batches.

We need an input parameter $\alpha > 0$. Define $\bar{y} = \lceil \log \log I \rceil$, which will serve as the upper bound for exploration in the inventory space.

We are now ready to present the LAIS algorithm, which repeats the following steps for each batch $i = 1, \dots, M$.

Step 1: Ordering during exploration periods. For $t \in \{t_i + 1, \dots, t_i + L\}$, set y_t as follows:

- (i) for $t = t_i + 1$, set $y_t = x_t$;
- (ii) for $t = t_i + 2, \dots, t_i + L$, set

$$y_t = \begin{cases} y_{t-1}, & \text{if } y_{t-1} > D_{t-1}^v; \\ \min \{ \lceil \max \{ y_{t-1}, 1 \} (1 + \alpha) \rceil, \bar{y} \}, & \text{otherwise.} \end{cases}$$

Step 2: Estimation using the most recent data set. The demand distribution is estimated as follows: $\hat{\mathbf{w}}_i = (\hat{w}_i^0, \hat{w}_i^1, \dots)$, where for $m = 0, 1, \dots$,

$$\hat{w}_i^m = \frac{1}{L} \sum_{t=t_i+1}^{t_i+L} \mathbb{1}(s_t = m).$$

Step 3: Minimizing the proxy cost function. Construct the proxy $Q(\hat{\mathbf{w}}_i, y)$, and then minimize it over $\mathcal{Y}$,

$$\max_{y \in \mathcal{Y}} Q(\hat{\mathbf{w}}_i, y).$$

Let the maximizer be denoted by $\hat{y}_i$.

Step 4: Setting inventory order-up-to levels for the exploitation phase. For $t = t_i + L + 1, \dots, t_i + L + I$, set the inventory order-up-to level to

$$y_t = \max\{x_t, \hat{y}_i\}.$$

As the algorithm shows, Step 1 focuses on exploration in the inventory space. Whenever inventory runs out, LAIS raises the order-up-to level for the next period by a certain percentage. The inventory level is increased to collect more demand information. But since demand can be unlimited, it is impossible to eliminate stockouts completely. Whenever stockout happens, $s_t = y_t < D_t^v$, which means the sales data that will be used for demand estimation suffer from downward biases. Step 2 estimates the densities using averages of sales samples from the current batch. LAIS does not use data from previous batches because the total number of change points is unknown, and it is totally possible that the demand distribution has already changed for the current batch, in which case sales data from previous batches become irrelevant and misleading. Step 3 minimizes the data-driven cost function with the true demand distribution replaced by $\hat{\mathbf{w}}_i$, and the resultant optimal solution $\hat{y}_i$ is implemented during every period of the exploitation phase in Step 4. But because of carryover inventories, $\hat{y}_i$ may not be met for some periods, in which case no new orders are placed and the inventory level stays as x_t .

4.2. Upper Bound Analysis

The theoretical upper bound of regret for the LAIS algorithm is given below.

THEOREM 2. *Under Assumption 1, the regret of LAIS is upper bounded by*

$$\mathcal{R}^{LAIS}(\Lambda, T) \leq 2 \max\{b, h\} \sqrt{T} (\log T) (\log \log T)^4.$$

Together with the lower bound in Theorem 1, it shows that the performance of LAIS is optimal up to the logarithmic terms. This is the first tight result for data-driven inventory control with censored

demand in a shifting environment. In the remainder of this section, we provide the key ideas in proving Theorem 2. Since the learning algorithm is based on estimation of demand distribution, an essential step is to obtain the convergence rates for their estimates. Proposition 3 below develops an upper bound for the estimation errors of the demand estimator $\hat{\mathbf{w}}_i$, and Proposition 4 will be used to evaluate the impact of the carry-over inventory constraint on the algorithm performance.

Let $C^1, \dots, C^V$ be the index of batches that contain periods $l^1, \dots, l^V$ (time periods when the demand changes). Note that some periods of $l^1, \dots, l^V$ may fall into the same batch, thus we allow $C^1, \dots, C^V$ to have repetitions and set $C = C^1 \cup \dots \cup C^V$. If a stage index $i \in C$, then the corresponding batch contains at least one demand change. If batch $i \notin C$, there is no demand change during this batch, in which case let $\mathcal{V}(i)$ denote the demand type over the batch i , let $\mathbf{w}^{\mathcal{V}(i)}$ denote the unique demand distribution in batch i , and let $y^{\mathcal{V}(i)*}$ denote the corresponding optimal order-up-to level. Similarly, for a given period t , let $\mathbf{w}^{\mathcal{V}(t)}$ be the parameters in period t , and the corresponding optimal order-up-to level is $y^{\mathcal{V}(t)*}$.

PROPOSITION 3. *Under Assumption 1, for any $i \notin C$ and any $y \in \mathcal{Y}$, when $I = \lceil \sqrt{T} \rceil$ is large enough we have,*

$$\mathbb{P}(|Q(\mathbf{w}^{\mathcal{V}(i)}, y) - Q(\hat{\mathbf{w}}_i, y)| \leq (3b+2)(\log \log I)^{-\frac{1}{2}}) \geq 1 - \frac{3(\log \log I)^{\frac{1}{4}}}{\theta_2 I^2}.$$

This proposition shows that for any $y \in \mathcal{Y}$, with a high probability, the objective value of y under the estimator $\hat{\mathbf{w}}_i$ is very close to that under the true parameter $\mathbf{w}^{\mathcal{V}(i)}$. This result indirectly demonstrates the estimation accuracy of $\hat{\mathbf{w}}_i$, and builds the confidence of using $\hat{\mathbf{w}}_i$ to evaluate a solution. Using $\hat{\mathbf{w}}_i$ in the place of $\mathbf{w}^{\mathcal{V}(i)}$, the corresponding optimal solution $\hat{y}_i$ should not be very far away from the true optimal solution under $\mathbf{w}^{\mathcal{V}(i)}$.

After obtaining estimators $\hat{\mathbf{w}}_i$, LAIS prescribes order-up-to levels $\hat{y}_i$ for the exploitation phase. Because of carry-over inventories, $\hat{y}_i$ may not always be achieved (Step 4 of LAIS), resulting in possible overshooting. The following proposition will be used to bound the profit loss due to overshooting.

PROPOSITION 4. *For any $v \in \{0, 1, \dots, V\}$ and large enough I , we have*

$$\mathbb{P}\left(\sum_{t=t_i+L+1}^{t_i+L+\log I \log \log I} D_t^v \geq \bar{y}\right) \geq 1 - \frac{1}{I^2}.$$

According to this proposition, with a high probability, the initial inventory x_{t_i+L+1} at the beginning of the exploitation phase will be consumed within the first $\log I \log \log I$ periods. After that, the order-up-to target $\hat{y}_i$ will be achieved in each period of the exploitation phase.

The proofs for Propositions 3 and 4 are relegated to the Appendix.

Proof of Theorem 2. For any $V, \bar{\mathbf{w}}, \bar{\mathbf{I}}$, one has,

$$\begin{aligned} \mathcal{Q}^\pi(V, \bar{\mathbf{I}}, \bar{\mathbf{w}}, T) - \tilde{\mathcal{Q}}(V, \bar{\mathbf{I}}, \bar{\mathbf{w}}, T) &= \sum_{v=0}^V \sum_{t=l^v}^{l^{v+1}-1} \mathbb{E}[Q(\mathbf{w}^v, y_t) - Q(\mathbf{w}^v, y^{v*})] \\ &\leq \sum_{i \in C} \sum_{t=t_i+1}^{t_i+L+I} \mathbb{E}[Q(\mathbf{w}^{\nu(t)}, y_t) - Q(\mathbf{w}^{\nu(t)}, y^{\nu(t)*})] \\ &\quad + \sum_{i \notin C} \sum_{t=t_i+1}^{t_i+L+I} \mathbb{E}[Q(\mathbf{w}^{\nu(i)}, y_t) - Q(\mathbf{w}^{\nu(i)}, y^{\nu(i)*})] \\ &\leq \sum_{i \in C} \sum_{t=t_i+1}^{t_i+L+I} \mathbb{E}[Q(\mathbf{w}^{\nu(t)}, y_t) - Q(\mathbf{w}^{\nu(t)}, y^{\nu(t)*})] \end{aligned} \quad (11)$$

$$+ \sum_{i \notin C} \left(\sum_{t=t_i+1}^{t_i+L} \mathbb{E}[Q(\mathbf{w}^{\nu(i)}, y_t) - Q(\mathbf{w}^{\nu(i)}, y^{\nu(i)*})] \right) \quad (12)$$

$$+ \sum_{t=t_i+L+1}^{t+L+I} \mathbb{E}[Q(\mathbf{w}^{\nu(i)}, \hat{y}_i) - Q(\mathbf{w}^{\nu(i)}, y^{\nu(i)*})] \quad (13)$$

$$+ \sum_{t=t_i+L+1}^{t+L+I} \mathbb{E}[|Q(\mathbf{w}^{\nu(i)}, \hat{y}_i) - Q(\mathbf{w}^{\nu(i)}, y_t)|] \Big). \quad (14)$$

The first inequality splits the total cost into batches either belonging to C or not. The second inequality further splits batches not in C into exploration periods and exploitation periods. The term in (11) stems from potential failures in detecting demand changes, and can be upper bounded by the total length of all the batches that contain demand changes. The term in (12) represents the increase in cost due to active exploration in the inventory space, and is upper bounded by the length of the exploration phase in every batch. The term in (13) depends on the estimation accuracy of $\hat{\mathbf{w}}_i$, and is bounded based on the result in Proposition 3. The term in (14) measures the performance loss due to overshooting, and is upper bounded based on Proposition 4. Summarizing all the analyses above, Theorem 2 can be proved. A formal proof is provided in the appendix.

5. Numerical Results

In this section, we conduct numerical experiments for LAIS. We measure the performance of a learning algorithm π by the percentage of relative regret defined as

$$\frac{\mathcal{R}^\pi(\Lambda, T)}{\mathcal{Q}^{\pi^*}(V, \bar{\mathbf{I}}, \bar{\mathbf{w}}, T)} \times 100\%,$$

where recall that $Q^{\pi^*}(V, \bar{\mathbf{I}}, \bar{\mathbf{w}}, T)$ is the optimal objective value under complete information. We test 24 problem scenarios, which will be explained in details shortly. For each scenario, we conduct 500 runs, compute the average, summarize and report the results in Figures 1 to 3.

The experiment setting is the following. We test lost-sale penalty cost $b \in \{1, 3, 5\}$ and holding cost $h \in \{1, 2\}$. We test the number of demand changes $V \in \{1, 2, 3, 4\}$. A combination of all the choices yields 24 scenarios in total. The set of potential demands Λ consists of 9 demand distributions as follows:

- poisson distribution with arrival rate $\lambda \in \{12, 14, 16\}$;
- binomial distributions with number of trials 30 and success probability $p \in \{2/5, 7/15, 8/15\}$;
- geometric distribution with success probability $p \in \{1/12, 1/14, 1/16\}$.

For each run, the time periods for demand change are randomly chosen with equal densities over $\{1, \dots, T\}$, and the new distribution the demand changes to is also randomly chosen with equal densities over the potential demand set Λ .

Figure 1 shows the average of regret over all the 24 scenarios tested, and the dotted curve represents the function of $0.175\sqrt{T}(\log T)^{1.4}$. This comparison shows that the performance of LAIS is pretty close to the rate of $O(\sqrt{T}(\log T)^{1.4})$, which is also consistent with our convergence result in Theorem 2.

We then average results of the 24 scenarios over $V = 1, 2, 3, 4$ and present the result in Figure 2. As it can be seen, the regret becomes larger when there are more demand changes, which is also very intuitive.

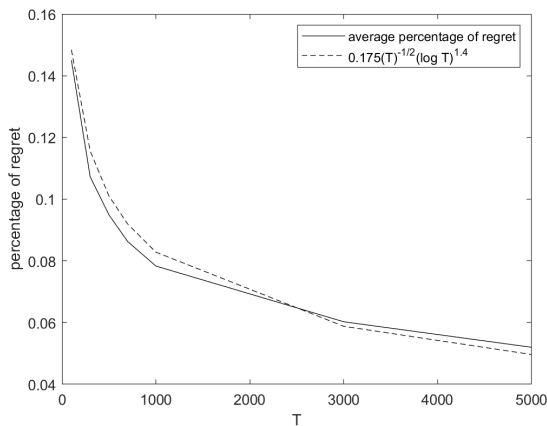


Figure 1 Overall average of regret

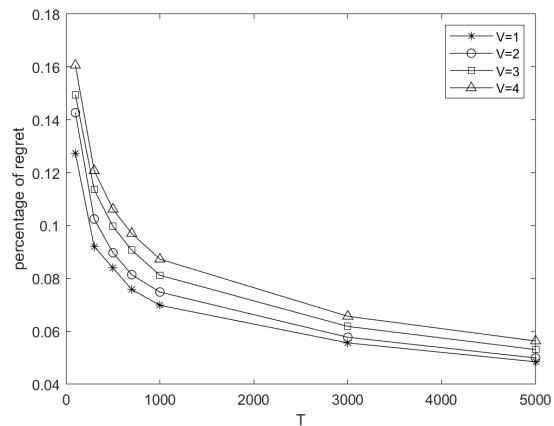


Figure 2 Comparisons of averages of regret for $V = 1, 2, 3, 4$

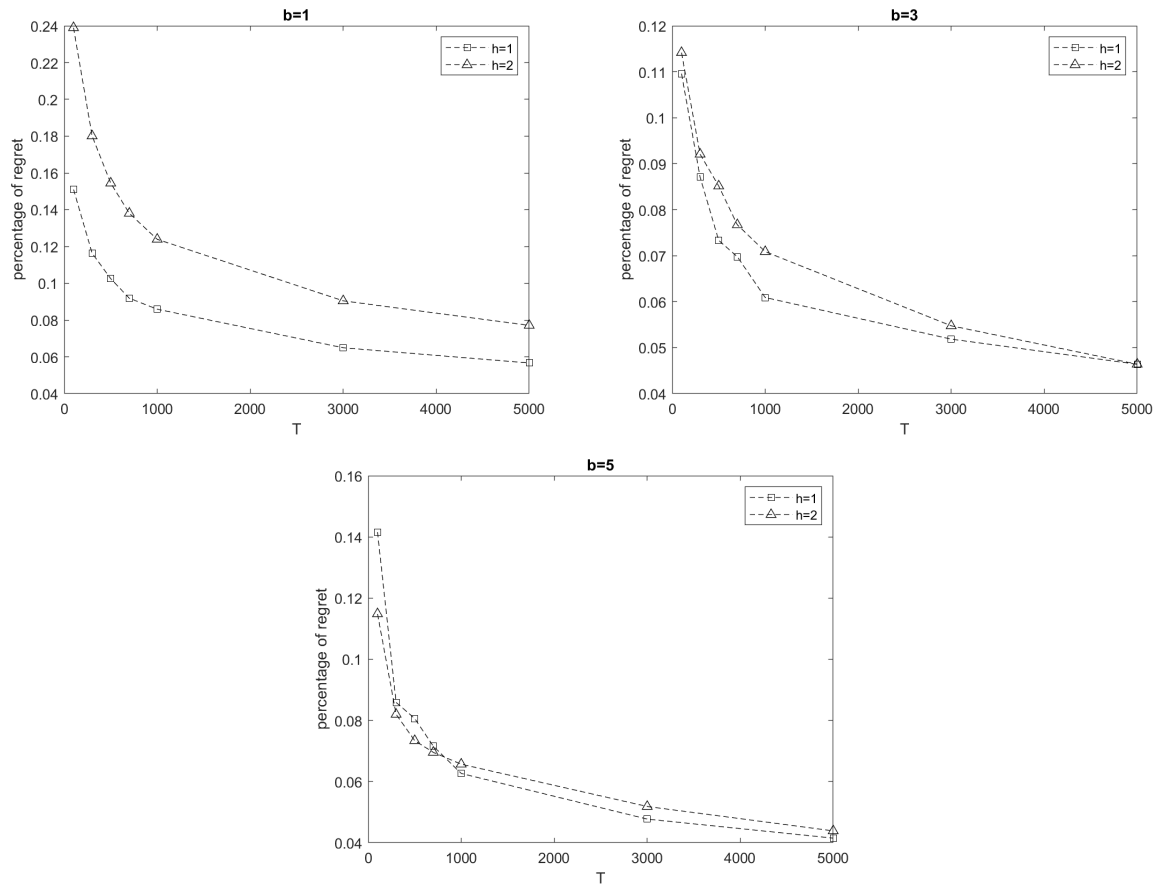


Figure 3 Comparisons of averages of regret for $h = 1$ and $h = 2$: top-left is under $b = 1$, top-right is under $b = 3$, bottom is under $b = 5$.

In Figure 3, we explore the effect of holding cost h . Recall that during the exploration periods of LAIS, the inventory level is intentionally raised to learn more demand information, creating higher holding cost. Therefore, one would expect the regret to increase in h . This conjecture is confirmed by the three comparisons in Figure 3. One sees that under any values of b , the regret is indeed higher for $h = 2$ than $h = 1$, but when the values of b changes from 1 to 3 then to 5, the gap between $h = 2$ and $h = 1$ becomes less and less significant.

6. Concluding Remarks

In this paper, we study the dynamic inventory control problem under nonparametric demand learning in a shifting demand environment. One salient feature is that neither the objective feedback (zeroth order feedback) nor the gradient feedback (first order feedback) is observable by the firm, therefore, approaches in the existing literature cannot be applied to solve this problem since they all require either of these two feedbacks. We provide the theoretical lower bound for this problem by analyzing two categories of policies: exploration-heavy policy and exploitation-heavy policy,

and prove that no matter which category an admissible policy falls into, the regret for it cannot be lower than $\Omega(\sqrt{T})$. Then we develop a data-driven algorithm (LAIS) that keeps updating the estimation of the demand distribution every $O(\sqrt{T})$ periods to offset the negative impact brought by the shifting environment. Since demands can be unbounded, it is not possible to completely eliminate stockouts, so the estimators developed in LAIS suffer from downward biases. Despite these factors, we prove that the convergence rate of LAIS matches the lower bound up to a $\log T$ term. This is the first tight result in data-driven inventory control under censored demand in a shifting environment.

In this paper, we only consider the most basic model in inventory control, which is the stochastic inventory control system with one echelon, no leadtime, and no setup cost. It will be interesting future directions to see whether it is possible to consider more complex models as mentioned above, that is, systems with multi-echelon, positive leadtime, and positive setup cost, and to develop learning algorithms with provably good performance guarantee for these models with shifting demands.

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Appendix

Note that proofs for Propositions 1 and 2 are based on the following problem instance. The support for the customer demand is $\{0, 1, 2\}$. Let $\Lambda = \{\mathbf{w}^0 = [3/5, 1/5, 1/5], \mathbf{w}^1 = [1/5, 1/5, 3/5]\}$. The number of demand changes $V = 1$, and the demand distribution switches from $\mathbf{w}^0$ to $\mathbf{w}^1$ at the beginning of period l^1 , $1 < l^1 \leq T$. Cost parameters $b = 1, h = 1$, and inventory decisions $y_t \in \{0, 1, 2\}$. Next we further reduce the complexity of the problem in two regards: First, assume the firm knows that the demand changes from $\mathbf{w}^0$ to $\mathbf{w}^1$ but does not know the time of change l^1 . Second, we remove the carry-over inventory constraint $y_t \geq (x_{t-1} - d_{t-1})^+$, so that the problem is reduced from the original constrained learning problem to an unconstrained problem. Any admissible policy for the original problem is also admissible for this case with reduced complexity, therefore, lower bound of regret for the latter is also a lower bound for the original problem with less information and more constraints. For $v = 0, 1$, the firm would like to solve the following single period problem,

$$\min_{y \in \mathcal{Y}} Q(\mathbf{w}^v, y) = \min_{y \in \mathcal{Y}} \{ \mathbb{E}[D^v - y]^+ + \mathbb{E}[y - D^v]^+ \}.$$

One sees that

$$Q(\mathbf{w}^0, 0) = \frac{3}{5}, \quad Q(\mathbf{w}^0, 1) = \frac{4}{5}, \quad Q(\mathbf{w}^0, 2) = \frac{7}{5},$$

and under $\mathbf{w}^0$, the optimal solution $y^{0*} = 0$.

On the other hand,

$$Q(\mathbf{w}^1, 0) = \frac{7}{5}, \quad Q(\mathbf{w}^1, 1) = \frac{4}{5}, \quad Q(\mathbf{w}^1, 2) = \frac{3}{5},$$

and under $\mathbf{w}^1$, the optimal solution $y^{1*} = 2$.

Under $\mathbf{w}^v$, $v = 0, 1$, when inventory order-up-to level is y , the distribution for sales quantity s is the following: for $y = 0$,

$$\mathbb{P}(0|y = 0) = 1;$$

for $y \geq 1$,

$$\mathbb{P}(s|y \geq 1) = \begin{cases} w^{vs} & \text{if } s \leq y - 1, \\ 1 - \sum_{m=1}^{y-1} w^{vm} & \text{if } s = y. \end{cases}$$

Divide the planning horizon $\{1, 2, \dots, T\}$ into batches of $\lceil \sqrt{T} \rceil$ periods, and define

$$N = \left\lceil \frac{T}{\lceil \sqrt{T} \rceil} \right\rceil, \text{ and } n_j = 1 + (j-1)\lceil \sqrt{T} \rceil, \text{ for } j = 1, \dots, N.$$

Here, N is the number of batches, and n_j is the first period of the j th batch. Let $\mathbb{P}_j^\pi$ be the distribution for sales quantities under policy π when the change in the demand distribution occurs at the beginning of period n_j .

Let $\mathcal{K}(\mathbb{P}_{j+1}^\pi, \mathbb{P}_j^\pi)$ denote the Kulback-Leibler (KL) divergence between $\mathbb{P}_{j+1}^\pi$ and $\mathbb{P}_j^\pi$, and is defined by

$$\begin{aligned} \mathcal{K}(\mathbb{P}_{j+1}^\pi, \mathbb{P}_j^\pi) &= \mathbb{E}_{j+1}^\pi \left[\log \frac{\mathbb{P}_{j+1}^\pi(s_t : 1 \leq t \leq T)}{\mathbb{P}_j^\pi(s_t : 1 \leq t \leq T)} \right] \\ &= \mathbb{E}_{j+1}^\pi \left[\log \frac{\mathbb{P}_{j+1}^\pi(s_T | \mathcal{H}_{T-1}) \cdots \mathbb{P}_{j+1}^\pi(s_2 | \mathcal{H}_1) \mathbb{P}_{j+1}^\pi(s_1)}{\mathbb{P}_j^\pi(s_T | \mathcal{H}_{T-1}) \cdots \mathbb{P}_j^\pi(s_2 | \mathcal{H}_1) \mathbb{P}_j^\pi(s_1)} \right] = \mathbb{E}_{j+1}^\pi \left[\log \prod_{t=1}^T \frac{\mathbb{P}_{j+1}^\pi(s_t | \mathcal{H}_{t-1})}{\mathbb{P}_j^\pi(s_t | \mathcal{H}_{t-1})} \right] \\ &= \mathbb{E}_{j+1}^\pi \left[\sum_{t=1}^T \log \frac{\mathbb{P}_{j+1}^\pi(s_t | \mathcal{H}_{t-1})}{\mathbb{P}_j^\pi(s_t | \mathcal{H}_{t-1})} \right] = \mathbb{E}_{j+1}^\pi \left[\sum_{t=n_j}^{n_{j+1}-1} \log \frac{\mathbb{P}_{\mathbf{w}^0}^\pi(s_t | \mathcal{H}_{t-1})}{\mathbb{P}_{\mathbf{w}^1}^\pi(s_t | \mathcal{H}_{t-1})} \right] \\ &= \sum_{t=n_j}^{n_{j+1}-1} \mathbb{E}_{j+1}^\pi \left[\log \frac{\mathbb{P}_{\mathbf{w}^0}^\pi(s_t | \mathcal{H}_{t-1})}{\mathbb{P}_{\mathbf{w}^1}^\pi(s_t | \mathcal{H}_{t-1})} \right] = \sum_{t=n_j}^{n_{j+1}-1} \mathbb{E}_{j+1}^\pi \left[\log \frac{\mathbb{P}_{\mathbf{w}^0}^\pi(s_t | y_t(\mathcal{H}_{t-1}))}{\mathbb{P}_{\mathbf{w}^1}^\pi(s_t | y_t(\mathcal{H}_{t-1}))} \right]. \end{aligned}$$

Proof of Theorem 1. Combing Propositions 1 and 2, Theorem 1 then follows. $\square$

Proof of Proposition 1. We first consider the case that $l^1 = n_N$. For any admissible policy π we have,

$$\begin{aligned} &\mathbb{E}_N^\pi [Q^\pi(V, \bar{\mathbf{I}}, \bar{\mathbf{w}}, T) - \tilde{Q}(V, \bar{\mathbf{I}}, \bar{\mathbf{w}}, T)] \\ &= \mathbb{E}_N^\pi \left[\sum_{t=1}^{n_N-1} [Q(\mathbf{w}^0, y_t(\mathcal{H}_{t-1})) - Q(\mathbf{w}^0, y^{0*})] + \sum_{t=n_N}^T [Q(\mathbf{w}^1, y_t(\mathcal{H}_{t-1})) - Q(\mathbf{w}^1, y^{1*})] \right] \\ &\geq \mathbb{E}_N^\pi \left[\sum_{t=1}^{n_N-1} [Q(\mathbf{w}^0, y_t(\mathcal{H}_{t-1})) - Q(\mathbf{w}^0, y^{0*})] \right] \\ &= \sum_{j=1}^{N-1} \mathbb{E}_N^\pi \left[\sum_{t=n_j}^{n_{j+1}-1} [Q(\mathbf{w}^0, y_t(\mathcal{H}_{t-1})) - Q(\mathbf{w}^0, y^{0*})] \right] \\ &\geq \sum_{j=1}^{N-1} \mathbb{E}_N^\pi \left[\sum_{t=n_j}^{n_{j+1}-1} \frac{1}{5} |y^{0*} - y_t(\mathcal{H}_{t-1})| \right], \end{aligned} \tag{15}$$

where the last inequality follows from $Q(\mathbf{w}^0, 0) - Q(\mathbf{w}^0, 1) = \frac{1}{5}$, and the fact that $Q(\mathbf{w}^0, y)$ is convex in y with maximizer $y^{0*} = 0$.

Recall that $\mathcal{K}(\mathbb{P}_{j+1}^\pi, \mathbb{P}_j^\pi)$ is given by

$$\mathcal{K}(\mathbb{P}_{j+1}^\pi, \mathbb{P}_j^\pi) = \sum_{t=n_j}^{n_{j+1}-1} \mathbb{E}_{j+1}^\pi \left[\log \frac{\mathbb{P}_{\mathbf{w}^0}(s_t | y_t(\mathcal{H}_{t-1}))}{\mathbb{P}_{\mathbf{w}^1}(s_t | y_t(\mathcal{H}_{t-1}))} \right] = \sum_{t=n_j}^{n_{j+1}-1} \mathbb{E}_{j+1}^\pi \left[\mathbb{E}_{j+1}^\pi \left[\log \frac{\mathbb{P}_{\mathbf{w}^0}(s_t | y_t(\mathcal{H}_{t-1}))}{\mathbb{P}_{\mathbf{w}^1}(s_t | y_t(\mathcal{H}_{t-1}))} \middle| y_t(\mathcal{H}_{t-1}) \right] \right],$$

and we analyze the following three cases.

Case (1): If $y_t(\mathcal{H}_{t-1}) = 0$,

$$\mathbb{E}_{j+1}^\pi \left[\log \frac{\mathbb{P}_{\mathbf{w}^0}(s_t | y_t(\mathcal{H}_{t-1}))}{\mathbb{P}_{\mathbf{w}^1}(s_t | y_t(\mathcal{H}_{t-1}))} \middle| y_t(\mathcal{H}_{t-1}) = 0 \right] = \log \frac{\mathbb{P}_{\mathbf{w}^0}(s_t = 0 | y_t(\mathcal{H}_{t-1}) = 0)}{\mathbb{P}_{\mathbf{w}^1}(s_t = 0 | y_t(\mathcal{H}_{t-1}) = 0)} = 0,$$

and

$$\frac{1}{5} |y^{0*} - y_t(\mathcal{H}_{t-1})| = 0,$$

which lead to

$$\mathbb{E}_{j+1}^\pi \left[\log \frac{\mathbb{P}_{\mathbf{w}^0}(s_t | y_t(\mathcal{H}_{t-1}))}{\mathbb{P}_{\mathbf{w}^1}(s_t | y_t(\mathcal{H}_{t-1}))} \middle| y_t(\mathcal{H}_{t-1}) \right] \leq \frac{1}{5} |y^{0*} - y_t(\mathcal{H}_{t-1})|.$$

Case (2): If $y_t(\mathcal{H}_{t-1}) = 1$,

$$\begin{aligned} & \mathbb{E}_{j+1}^\pi \left[\log \frac{\mathbb{P}_{\mathbf{w}^0}(s_t | y_t(\mathcal{H}_{t-1}))}{\mathbb{P}_{\mathbf{w}^1}(s_t | y_t(\mathcal{H}_{t-1}))} \middle| y_t(\mathcal{H}_{t-1}) = 1 \right] \\ &= \mathbb{P}_{\mathbf{w}^0}(s_t = 0 | y_t(\mathcal{H}_{t-1}) = 1) \log \frac{\mathbb{P}_{\mathbf{w}^0}(s_t = 0 | y_t(\mathcal{H}_{t-1}) = 1)}{\mathbb{P}_{\mathbf{w}^1}(s_t = 0 | y_t(\mathcal{H}_{t-1}) = 1)} \\ & \quad + \mathbb{P}_{\mathbf{w}^0}(s_t = 1 | y_t(\mathcal{H}_{t-1}) = 1) \log \frac{\mathbb{P}_{\mathbf{w}^0}(s_t = 1 | y_t(\mathcal{H}_{t-1}) = 1)}{\mathbb{P}_{\mathbf{w}^1}(s_t = 1 | y_t(\mathcal{H}_{t-1}) = 1)} \\ &= \frac{3}{5} \log \frac{3/5}{1/5} + \frac{2}{5} \log \frac{2/5}{4/5} = \frac{3}{5} \log 3 - \frac{2}{5} \log 2. \end{aligned}$$

and

$$\frac{1}{5} |y^{0*} - y_t(\mathcal{H}_{t-1})| = \frac{1}{5},$$

which leads to

$$\mathbb{E}_{j+1}^\pi \left[\log \frac{\mathbb{P}_{\mathbf{w}^0}(s_t | y_t(\mathcal{H}_{t-1}))}{\mathbb{P}_{\mathbf{w}^1}(s_t | y_t(\mathcal{H}_{t-1}))} \middle| y_t(\mathcal{H}_{t-1}) \right] \leq \frac{1}{5} |y^{0*} - y_t(\mathcal{H}_{t-1})|.$$

Case (3): If $y_t(\mathcal{H}_{t-1}) = 2$,

$$\mathbb{E}_{j+1}^\pi \left[\log \frac{\mathbb{P}_{\mathbf{w}^0}(s_t | y_t(\mathcal{H}_{t-1}))}{\mathbb{P}_{\mathbf{w}^1}(s_t | y_t(\mathcal{H}_{t-1}))} \middle| y_t(\mathcal{H}_{t-1}) = 2 \right]$$

$$\begin{aligned}
&= \mathbb{P}_{\mathbf{w}^0}(s_t = 0 | y_t(\mathcal{H}_{t-1}) = 2) \log \frac{\mathbb{P}_{\mathbf{w}^0}(s_t = 0 | y_t(\mathcal{H}_{t-1}) = 2)}{\mathbb{P}_{\mathbf{w}^1}(s_t = 0 | y_t(\mathcal{H}_{t-1}) = 2)} \\
&\quad + \mathbb{P}_{\mathbf{w}^0}(s_t = 1 | y_t(\mathcal{H}_{t-1}) = 2) \log \frac{\mathbb{P}_{\mathbf{w}^0}(s_t = 1 | y_t(\mathcal{H}_{t-1}) = 2)}{\mathbb{P}_{\mathbf{w}^1}(s_t = 1 | y_t(\mathcal{H}_{t-1}) = 2)} \\
&\quad + \mathbb{P}_{\mathbf{w}^0}(s_t = 2 | y_t(\mathcal{H}_{t-1}) = 2) \log \frac{\mathbb{P}_{\mathbf{w}^0}(s_t = 2 | y_t(\mathcal{H}_{t-1}) = 2)}{\mathbb{P}_{\mathbf{w}^1}(s_t = 2 | y_t(\mathcal{H}_{t-1}) = 2)} \\
&= \frac{3}{5} \log \frac{3/5}{1/5} + \frac{1}{5} \log \frac{1/5}{1/5} + \frac{1}{5} \log \frac{1/5}{3/5} = \frac{2}{5} \log 3.
\end{aligned}$$

and

$$\frac{1}{5} |y^{0*} - y_t(\mathcal{H}_{t-1})| = \frac{2}{5},$$

and this again leads to

$$\mathbb{E}_{j+1}^{\pi} \left[\log \frac{\mathbb{P}_{\mathbf{w}^0}(s_t | y_t(\mathcal{H}_{t-1}))}{\mathbb{P}_{\mathbf{w}^1}(s_t | y_t(\mathcal{H}_{t-1}))} \middle| y_t(\mathcal{H}_{t-1}) \right] \leq \frac{1}{5} |y^{0*} - y_t(\mathcal{H}_{t-1})|.$$

Summarizing cases (1)-(3), we conclude that for any $y_t(\mathcal{H}_{t-1}) \in \{0, 1, 2\}$,

$$\mathbb{E}_{j+1}^{\pi} \left[\log \frac{\mathbb{P}_{\mathbf{w}^0}(s_t | y_t(\mathcal{H}_{t-1}))}{\mathbb{P}_{\mathbf{w}^1}(s_t | y_t(\mathcal{H}_{t-1}))} \middle| y_t(\mathcal{H}_{t-1}) \right] \leq \frac{1}{5} |y^{0*} - y_t(\mathcal{H}_{t-1})|,$$

and therefore

$$\mathbb{E}_{j+1}^{\pi} \left[\mathbb{E}_{j+1}^{\pi} \left[\log \frac{\mathbb{P}_{\mathbf{w}^0}(s_t | y_t(\mathcal{H}_{t-1}))}{\mathbb{P}_{\mathbf{w}^1}(s_t | y_t(\mathcal{H}_{t-1}))} \middle| y_t(\mathcal{H}_{t-1}) \right] \right] \leq \mathbb{E}_{j+1}^{\pi} \left[\frac{1}{5} |y^{0*} - y_t(\mathcal{H}_{t-1})| \right] = \mathbb{E}_N^{\pi} \left[\frac{1}{5} |y^{0*} - y_t(\mathcal{H}_{t-1})| \right],$$

where the last equality is true because $t \leq n_{j+1} - 1$ and $j \leq N - 1$.

The above analyses imply that

$$1 < \mathcal{K}(\mathbb{P}_{j+1}^{\pi}, \mathbb{P}_j^{\pi}) \leq \mathbb{E}_N^{\pi} \left[\sum_{t=n_j}^{n_{j+1}-1} \frac{1}{5} |y^{0*} - y_t(\mathcal{H}_{t-1})| \right].$$

Therefore, based on (15), one has

$$\mathcal{R}^{\pi}(\Lambda, T) \geq \mathbb{E}_N^{\pi} [\mathcal{Q}^{\pi}(V, \bar{\mathbf{l}}, \bar{\mathbf{w}}, \bar{\mathbf{a}}, T) - \mathcal{Q}^{\pi^*}(V, \bar{\mathbf{l}}, \bar{\mathbf{w}}, \bar{\mathbf{a}}, T)] > N - 1.$$

Proposition 1 is thus proved. $\square$

Proof of Proposition 2. Consider an admissible policy π such that $\min_{1 \leq j \leq N-1} \mathcal{K}(\mathbb{P}_{j+1}^{\pi}, \mathbb{P}_j^{\pi}) \leq 1$, and let j_0 denote a index such that $\mathcal{K}(\mathbb{P}_{j_0+1}^{\pi}, \mathbb{P}_{j_0}^{\pi}) \leq 1$. Consider the following two hypotheses:

$$H_0: \quad l^1 = n_{j_0} \qquad H_1: \quad l^1 = n_{j_0+1}.$$

Let ψ be a test, i.e., a mapping from information during periods $\{1, \dots, T\}$ into $\{0, 1\}$: $\psi = 0$ denotes “ H_0 is true”, and $\psi = 1$ denotes “ H_1 is true”. By Tsybakov (2004, Theorem 2.2),

$$\inf_{\psi} \max \{ \mathbb{P}_{j_0}^{\pi}(\psi = 1), \mathbb{P}_{j_0+1}^{\pi}(\psi = 0) \} \geq \frac{1}{4e}. \quad (16)$$

For a moment assume

$$\max_{k=j_0, j_0+1} \mathbb{E}_k^{\pi} \left[\sum_{t=1}^{n_k-1} |y^{0*} - y_t(\mathcal{H}_{t-1})| + \sum_{t=n_k}^T |y^{1*} - y_t(\mathcal{H}_{t-1})| \right] \leq \frac{1}{8e} \lceil \sqrt{T} \rceil, \quad (17)$$

and consider the following test ψ :

$$\psi = \begin{cases} 0, & \text{if } \sum_{t=n_{j_0}}^{n_{j_0+1}-1} |y^{0*} - y_t(\mathcal{H}_{t-1})| > \lceil \sqrt{T} \rceil; \\ 1, & \text{if } \sum_{t=n_{j_0}}^{n_{j_0+1}-1} |y^{0*} - y_t(\mathcal{H}_{t-1})| \leq \lceil \sqrt{T} \rceil. \end{cases}$$

Next one has

$$\begin{aligned} \mathbb{P}_{j_0+1}^{\pi}(\psi = 0) &= \mathbb{P}_{j_0+1}^{\pi} \left(\sum_{t=n_{j_0}}^{n_{j_0+1}-1} |y^{0*} - y_t(\mathcal{H}_{t-1})| > \lceil \sqrt{T} \rceil \right) \leq \frac{1}{\lceil \sqrt{T} \rceil} \mathbb{E}_{j_0+1}^{\pi} \left[\sum_{t=n_{j_0}}^{n_{j_0+1}-1} |y^{0*} - y_t(\mathcal{H}_{t-1})| \right] \\ &\leq \frac{1}{\lceil \sqrt{T} \rceil} \mathbb{E}_{j_0+1}^{\pi} \left[\sum_{t=1}^{n_{j_0+1}-1} |y^{0*} - y_t(\mathcal{H}_{t-1})| + \sum_{t=n_{j_0+1}}^T |y^{1*} - y_t(\mathcal{H}_{t-1})| \right] \\ &\leq \frac{1}{\lceil \sqrt{T} \rceil} \frac{1}{8e} \lceil \sqrt{T} \rceil = \frac{1}{8e}, \end{aligned} \quad (18)$$

where the first inequality follows from Markov inequality, and the third inequality follows from (17).

On the other hand, one has

$$\begin{aligned} \mathbb{P}_{j_0}^{\pi}(\psi = 1) &= \mathbb{P}_{j_0}^{\pi} \left(\sum_{t=n_{j_0}}^{n_{j_0+1}-1} |y^{0*} - y_t(\mathcal{H}_{t-1})| \leq \lceil \sqrt{T} \rceil \right) \leq \mathbb{P}_{j_0}^{\pi} \left(\sum_{t=n_{j_0}}^{n_{j_0+1}-1} |y^{1*} - y_t(\mathcal{H}_{t-1})| \geq 2\lceil \sqrt{T} \rceil - \lceil \sqrt{T} \rceil \right) \\ &\leq \frac{1}{\lceil \sqrt{T} \rceil} \mathbb{E}_{j_0}^{\pi} \left[\sum_{t=n_{j_0}}^{n_{j_0+1}-1} |y^{1*} - y_t(\mathcal{H}_{t-1})| \right] \\ &\leq \frac{1}{\lceil \sqrt{T} \rceil} \mathbb{E}_{j_0}^{\pi} \left[\sum_{t=1}^{n_{j_0}-1} |y^{0*} - y_t(\mathcal{H}_{t-1})| + \sum_{t=n_{j_0}}^T |y^{1*} - y_t(\mathcal{H}_{t-1})| \right] \leq \frac{1}{8e}, \end{aligned} \quad (19)$$

where the first inequality follows from $|y^{0*} - y^{1*}| = 2$, the second inequality follows from Markov inequality, and the last inequality follows from (17).

Combing (18) and (19) presents a conflict of (16). Therefore, the assumption in (17) cannot hold, implying that

$$\max_{1 < l^1 \leq T} \mathbb{E}^\pi \left[\sum_{t=1}^{l^1-1} |y^{0*} - y_t(\mathcal{H}_{t-1})| + \sum_{t=l^1}^T |y^{1*} - y_t(\mathcal{H}_{t-1})| \right] > \frac{1}{8e} \lceil \sqrt{T} \rceil,$$

where inside the max operator, it is the expected deviation from the optimal order-up-to level if the demand changes at the beginning of period l^1 . This leads to

$$\begin{aligned} & \max_{1 < l^1 \leq T} \mathbb{E}^\pi \left[\sum_{t=1}^{l^1-1} (Q(\mathbf{w}^0, y_t(\mathcal{H}_{t-1})) - Q(\mathbf{w}^0, y^{0*})) + \sum_{t=l^1}^T (Q(\mathbf{w}^1, y_t(\mathcal{H}_{t-1})) - Q(\mathbf{w}^1, y^{1*})) \right] \\ & \geq \frac{1}{8e} \lceil \sqrt{T} \rceil \frac{1}{5} = \frac{1}{40e} \lceil \sqrt{T} \rceil, \end{aligned}$$

and the last inequality is true because

$$\min \{Q(\mathbf{w}^0, 0) - Q(\mathbf{w}^0, 1), Q(\mathbf{w}^1, 2) - Q(\mathbf{w}^1, 1)\} = \frac{1}{5},$$

$Q(\mathbf{w}^0, y)$ is concave in y with $y^{0*} = 0$, and $Q(\mathbf{w}^1, y)$ is concave in y with $y^{1*} = 2$.

Proposition 2 is thus proved. $\square$

Proof of Proposition 3. If a batch index $i \notin C$, there is a unique demand distribution for the whole batch, which we use $\mathcal{V}(i)$ to denote the type demand and $\mathbf{w}^{\mathcal{V}(i)}$ to denote the corresponding demand distribution with $\mathbf{w}^{\mathcal{V}(i)} = (w^{\mathcal{V}(i)m}, \text{ for } m = 0, 1, \dots)$. Let $\hat{D}_i$ be the random variable corresponding to $\hat{\mathbf{w}}_i$.

Let $\lceil (\log \log I)^{1/4} / \theta_2 \rceil = \bar{m}$. Since $\bar{y} \sim \mathcal{O}(\log \log I)$, we have $\bar{m} < \bar{y}$. For any $y \in \mathcal{Y}$, we compare $Q(\mathbf{w}^{\mathcal{V}(i)m}, y)$ and $Q(\hat{\mathbf{w}}_i, y)$ as the following.

$$\begin{aligned} & |Q(\mathbf{w}^{\mathcal{V}(i)m}, y) - Q(\hat{\mathbf{w}}_i, y)| \\ & = |b\mathbb{E}[(D^{\mathcal{V}(i)} - y)^+] + h\mathbb{E}[(y - D^{\mathcal{V}(i)})^+] - b\mathbb{E}[(\hat{D}_i - y)^+] - h\mathbb{E}[(y - \hat{D}_i)^+]| \\ & \leq h|\mathbb{E}[(y - D^{\mathcal{V}(i)})^+] - \mathbb{E}[(y - \hat{D}_i)^+]| + b|\mathbb{E}[(D^{\mathcal{V}(i)} - y)^+] - \mathbb{E}[(\hat{D}_i - y)^+]| \\ & \leq h \sum_{m=0}^y (y - m)|w^{\mathcal{V}(i)m} - \hat{w}_i^m| + b \sum_{m=y}^{+\infty} (m - y)|w^{\mathcal{V}(i)m} - \hat{w}_i^m| \\ & \leq h \sum_{m=0}^{\bar{y}} \bar{y}|w^{\mathcal{V}(i)m} - \hat{w}_i^m| + b \sum_{m=0}^{+\infty} m|w^{\mathcal{V}(i)m} - \hat{w}_i^m| \\ & \leq h \sum_{m=0}^{\bar{y}} \bar{y}|w^{\mathcal{V}(i)m} - \hat{w}_i^m| + b \sum_{m=0}^{\bar{m}} m|w^{\mathcal{V}(i)m} - \hat{w}_i^m| + b \sum_{m=\bar{m}+1}^{\bar{y}} m|w^{\mathcal{V}(i)m} - \hat{w}_i^m| + b \sum_{m=\bar{y}+1}^{\infty} m|w^{\mathcal{V}(i)m} - \hat{w}_i^m| \end{aligned}$$

$$\leq (h+b)\bar{m} \sum_{m=0}^{\bar{m}} |w^{\mathcal{V}(i)m} - \hat{w}_i^m| + b\bar{y} \sum_{m=\bar{m}+1}^{\bar{y}} |w^{\mathcal{V}(i)m} - \hat{w}_i^m| + b \sum_{m=\bar{y}+1}^{\infty} mw^{\mathcal{V}(i)m}. \quad (20)$$

Next we will provide upper bound for each of the three terms in (20). The first term is bounded as follows.

$$\begin{aligned} & \sum_{m=0}^{\bar{m}} |w^{\mathcal{V}(i)m} - \hat{w}_i^m| \\ &= \sum_{m=0}^{\bar{m}} \left| \frac{1}{L} \sum_{t=t_i+1}^{t_i+L} \mathbb{1}(s_t = m) - w^{\mathcal{V}(i)m} \right| \\ &\leq \sum_{m=0}^{\bar{m}} \left| \frac{1}{L} \sum_{t=t_i+1}^{t_i+L} \mathbb{1}(s_t = m) - \frac{1}{L} \sum_{t=t_i+1}^{t_i+L} \mathbb{1}(D_t^{\mathcal{V}(i)} = m) \right| + \sum_{m=0}^{\bar{m}} \left| \frac{1}{L} \sum_{t=t_i+1}^{t_i+L} \mathbb{1}(D_t^{\mathcal{V}(i)} = m) - w^{\mathcal{V}(i)m} \right| \\ &\leq \frac{1}{L} \sum_{t=t_i+1}^{t_i+L} \sum_{m=0}^{\bar{m}} |\mathbb{1}(s_t = m) - \mathbb{1}(D_t^{\mathcal{V}(i)} = m)| \end{aligned} \quad (21)$$

$$+ \sum_{m=0}^{\bar{m}} \left| \frac{1}{L} \sum_{t=t_i+1}^{t_i+L} \mathbb{1}(D_t^{\mathcal{V}(i)} = m) - w^{\mathcal{V}(i)m} \right|. \quad (22)$$

We first consider (21). For any $t \in \{t_i + 1, \dots, t_i + L\}$, we have

$$\sum_{m=0}^{\bar{m}} |\mathbb{1}(s_t = m) - \mathbb{1}(D_t^{\mathcal{V}(i)} = m)| \leq 2, \quad (23)$$

and if

$$\sum_{m=0}^{\bar{m}} |\mathbb{1}(s_t = m) - \mathbb{1}(D_t^{\mathcal{V}(i)} = m)| \neq 0,$$

then one has $s_t < D_t^{\mathcal{V}(i)}$. Note that $s_t < D_t^{\mathcal{V}(i)}$ happens when $y_t < D_t^{\mathcal{V}(i)}$ and $s_t = y_t$. Based on Step 1 of LAIS, to reach $\bar{m}$, the inventory level is raised by at most $\bar{M}$ times (which means there are at most $\bar{M}$ times of stockouts) before y_t reaches $\bar{m}$, where $\bar{M} \sim \mathcal{O}(\log \bar{m}) \sim \mathcal{O}(\log \log \log I)$. Using (23), we obtain

$$\sum_{t=t_i+1}^{t_i+L} \sum_{m=0}^{\bar{m}} |\mathbb{1}(s_t = m) - \mathbb{1}(D_t^{\mathcal{V}(i)} = m)| \leq 2\bar{M}. \quad (24)$$

When I is large enough, we have $2\bar{M} \leq \log \log I$. Hence applying (24), the term in (21) is upper bounded as

$$\frac{1}{L} \sum_{t=t_i+1}^{t_i+L} \sum_{m=0}^{\bar{m}} |\mathbb{1}(s_t = m) - \mathbb{1}(D_t^{\mathcal{V}(i)} = m)| \leq (\log I)^{-1} (\log \log I)^{-1}. \quad (25)$$

The term in (22) is bounded as follows. First of all, for any $m = 0, 1, \dots, \bar{m}$, notice that

$$\mathbb{E}[\mathbb{1}(D_t^{\mathcal{V}(i)} = m)] = w^{\mathcal{V}(i)m}.$$

By Hoeffding inequality, we have

$$\mathbb{P}\left(\left|\frac{1}{L} \sum_{t=t_i+1}^{t_i+L} \mathbb{1}(D_t^{\mathcal{V}(i)} = m) - w^{\mathcal{V}(i)m}\right| \leq (\log \log I)^{-3/2}\right) \geq 1 - \frac{2}{I^2},$$

which implies

$$\mathbb{P}\left(\sum_{m=0}^{\bar{m}} \left|\frac{1}{L} \sum_{t=t_i+1}^{t_i+L} \mathbb{1}(D_t^{\mathcal{V}(i)} = m) - w^{\mathcal{V}(i)m}\right| \leq (\log \log I)^{-3/2} \bar{m}\right) \geq 1 - \frac{2\bar{m}}{I^2}.$$

Therefore (22) can be bounded as

$$\mathbb{P}\left(\sum_{m=0}^{\bar{m}} \left|\frac{1}{L} \sum_{t=t_i+1}^{t_i+L} \mathbb{1}(D_t^{\mathcal{V}(i)} = m) - w^{\mathcal{V}(i)m}\right| \leq \theta_2^{-1} (\log \log I)^{-5/4}\right) \geq 1 - \frac{2(\log \log I)^{\frac{1}{4}}}{\theta_2 I^2}. \quad (26)$$

Plugging (25) and (26) into (21) and (22), one has that

$$\mathbb{P}\left(\sum_{m=0}^{\bar{m}} |w^{\mathcal{V}(i)m} - \hat{w}_i^m| \leq 2\theta_2^{-1} (\log \log I)^{-5/4}\right) \geq 1 - \frac{2(\log \log I)^{\frac{1}{4}}}{\theta_2 I^2},$$

which yields that the first term in (20) is upper bounded by

$$\mathbb{P}\left((h+b)\bar{m} \sum_{m=0}^{\bar{m}} |w^{\mathcal{V}(i)m} - \hat{w}_i^m| \leq 2(h+b)\theta_2^{-2} (\log \log I)^{-1}\right) \geq 1 - \frac{2(\log \log I)^{\frac{1}{4}}}{\theta_2 I^2}. \quad (27)$$

The second term in (20) is bounded as follows.

$$\begin{aligned} & \sum_{m=\bar{m}+1}^{\bar{y}} |w^{\mathcal{V}(i)m} - \hat{w}_i^m| \\ & \leq \sum_{m=\bar{m}+1}^{\bar{y}} w^{\mathcal{V}(i)m} \end{aligned} \quad (28)$$

$$+ \sum_{m=\bar{m}+1}^{\bar{y}} \hat{w}_i^m. \quad (29)$$

The term in (28) is bounded by Assumption 1(1) as follows:

$$\begin{aligned}
 \sum_{m=\bar{m}+1}^{\bar{y}} w^{\mathcal{V}(i)m} &\leq \sum_{m=\bar{m}+1}^{+\infty} w^{\mathcal{V}(i)m} \leq \sum_{m=\bar{m}+1}^{+\infty} \theta_1 e^{-\theta_2 m} = \frac{\theta_1 e^{-\theta_2}}{1 - e^{-\theta_2}} e^{-(\log \log I)^{1/4}} \\
 &\leq \frac{\theta_1 e^{-\theta_2}}{1 - e^{-\theta_2}} e^{-2 \log \log \log I} \\
 &= \frac{\theta_1 e^{-\theta_2}}{1 - e^{-\theta_2}} (\log \log I)^{-2}.
 \end{aligned} \tag{30}$$

To bound the term in (29), one has

$$\begin{aligned}
 \sum_{m=\bar{m}+1}^{\bar{y}} \hat{w}_i^m &= \sum_{m=\bar{m}+1}^{\bar{y}} \frac{1}{L} \sum_{t=t_i+1}^{t_i+L} \mathbb{1}(s_t = m) \\
 &= \frac{1}{L} \sum_{t=t_i+1}^{t_i+L} \mathbb{1}(s_t \geq \bar{m} + 1) \\
 &\leq \frac{1}{L} \sum_{t=t_i+1}^{t_i+L} \mathbb{1}\left(D_t^{\mathcal{V}(i)} \geq \bar{m} + 1\right).
 \end{aligned} \tag{31}$$

By Hoeffding's inequality, one has

$$\mathbb{P}\left(\left|\frac{1}{L} \sum_{t=t_i+1}^{t_i+L} \mathbb{1}\left(D_t^{\mathcal{V}(i)} \geq \bar{m} + 1\right) - \mathbb{P}\left(D_t^{\mathcal{V}(i)} \geq \bar{m} + 1\right)\right| \leq (\log \log I)^{-3/2}\right\} \geq 1 - \frac{2}{I^2},$$

and also

$$\mathbb{P}\left(D_t^{\mathcal{V}(i)} \geq \bar{m} + 1\right) = \sum_{m=\bar{m}+1}^{+\infty} w^{\mathcal{V}(i)m} \leq \frac{\theta_1 e^{-\theta_2}}{1 - e^{-\theta_2}} (\log \log I)^{-2},$$

where the inequality follows from (30).

Therefore,

$$\mathbb{P}\left(\frac{1}{L} \sum_{t=t_i+1}^{t_i+L} \mathbb{1}\left(D_t^{\mathcal{V}(i)} \geq \bar{m} + 1\right) \leq 2(\log \log I)^{-3/2}\right) \geq 1 - \frac{2}{I^2}. \tag{32}$$

From (32) and (31), we obtain that the term in (29) satisfies

$$\mathbb{P}\left(\sum_{m=\bar{m}+1}^{\bar{y}} \hat{w}_i^m \leq 2(\log \log I)^{-3/2}\right) \geq 1 - \frac{2}{I^2}. \tag{33}$$

Combining (30) and (33), the second term in (20) can be bounded as

$$\mathbb{P}(b\bar{y} \sum_{m=\bar{m}+1}^{\bar{y}} |w^{\mathcal{V}(i)m} - \hat{w}_i^m| \leq 3b(\log \log I)^{-1/2}) \geq 1 - \frac{2}{I^2}. \quad (34)$$

The third term in (20) is bounded as follows.

$$\begin{aligned} b \sum_{m=\bar{y}+1}^{\infty} m w^{\mathcal{V}(i)m} &\leq b\theta_1 \sum_{m=\bar{y}+1}^{\infty} m e^{-\theta_2 m} \\ &= b\theta_1 e^{-\theta_2(\bar{y}+1)} \left(\frac{\bar{y}+1}{1-e^{-\theta_2}} + \frac{e^{-\theta_2}}{(1-e^{-\theta_2})^2} \right) \\ &\leq b\theta_1 e^{-\theta_2 \log \log I} \left(\frac{\log \log I}{1-e^{-\theta_2}} + \frac{e^{-\theta_2}}{(1-e^{-\theta_2})^2} \right). \end{aligned} \quad (35)$$

Combine (27), (34) and (35), and replace them in (20), one has that for any $y \in \mathcal{Y}$,

$$\mathbb{P}(|Q(\mathbf{w}^{\mathcal{V}(i)}, y) - Q(\hat{\mathbf{w}}_i, y)| \leq (3b+2)(\log \log I)^{-\frac{1}{2}}) \geq 1 - \frac{3(\log \log I)^{\frac{1}{4}}}{\theta_2 I^2}.$$

Proof of Proposition 3 is thus complete. $\square$

Proof of Proposition 4. First, let D^v , $v = 0, \dots, V$, be the generic form for initial demand D_t^v , then by Lemma A1, one has

$$\mathbb{P}\left(\sum_{t=t_i+L+1}^{t_i+L+\log I \log \log I} D_t^v - \log I \log \log I \mathbb{E}[D^v] \geq -2\sqrt{2}\sigma \log I \sqrt{\log \log I}\right) \geq 1 - \frac{1}{I^2},$$

where σ is the standard deviation of D^v . When I is large enough, $1/2 \log I \log \log I \mathbb{E}[D^v] \geq 2\sqrt{2}\sigma \log I \sqrt{\log \log I}$, then one has

$$\mathbb{P}\left(\sum_{t=t_i+L+1}^{t_i+L+\log I \log \log I} D_t^v \geq \frac{1}{2} \log I \log \log I \mathbb{E}[D^v]\right) \geq 1 - \frac{1}{I^2}. \quad (36)$$

When I is large enough, $\frac{1}{2} \log I \log \log I \mathbb{E}[D^v] > \bar{y}$, therefore one has

$$\mathbb{P}\left(\sum_{t=t_i+L+1}^{t_i+L+\log I \log \log I} D_t^v \geq \bar{y}\right) \geq 1 - \frac{1}{I^2}.$$

Proposition 4 is thus proved. $\square$

Proof of Theorem 2. Let $C^1, \dots, C^V$ be the batches that contain periods $l^1, \dots, l^V$. Note that some periods of $l^1, \dots, l^V$ may fall into the same batch, thus we allow $C^1, \dots, C^V$ to have repetitions and set $C = C^1 \cup \dots \cup C^V$. If a batch index $i \in C$, then the corresponding stage contains at least

one shifting period. If an stage index $i \notin C$, then the i th stage has a uniform set of parameters $\mathbf{w}^{\nu(i)}$ for every period, for which the optimal solution is $y^{\nu(i)*}$. In case $Q(\mathbf{w}^{\nu(i)}, y)$ has more than one minimizer, let $\mathcal{Y}^{\nu(i)*}$ be the set of minimizers and $y^{\nu(i)*}$ be the smallest element.

For t , let $\mathbf{w}^{\nu(t)}$ be the parameters in period t , and the corresponding optimal order-up-to level is $y^{\nu(t)*}$. Similarly, for i , let $\mathbf{w}^{\nu(i)}$ be the parameters in batch i , and the corresponding optimal order-up-to level is $y^{\nu(i)*}$. Then for any $V, \bar{\mathbf{w}}, \bar{\mathbf{I}}$, one has,

$$\begin{aligned} \mathcal{Q}^\pi(V, \bar{\mathbf{I}}, \bar{\mathbf{w}}, T) - \tilde{\mathcal{Q}}(V, \bar{\mathbf{I}}, \bar{\mathbf{w}}, T) &= \mathbb{E} \left[\sum_{v=0}^V \sum_{t=l^v}^{l^{v+1}-1} Q(\mathbf{w}^v, y_t) - Q(\mathbf{w}^v, y^{v*}) \right] \\ &\leq \sum_{i \in C} \sum_{t=t_i+1}^{t_i+L+I} \mathbb{E} [Q(\mathbf{w}^{\nu(t)}, y_t) - Q(\mathbf{w}^{\nu(t)}, y^{\nu(t)*})] \\ &\quad + \sum_{i \notin C} \sum_{t=t_i+1}^{t_i+L+I} \mathbb{E} [Q(\mathbf{w}^{\nu(i)}, y_t) - Q(\mathbf{w}^{\nu(i)}, y^{\nu(i)*})] \\ &\leq \sum_{i \in C} \sum_{t=t_i+1}^{t_i+L+I} \mathbb{E} [Q(\mathbf{w}^{\nu(t)}, y_t) - Q(\mathbf{w}^{\nu(t)}, y^{\nu(t)*})] \end{aligned} \quad (37)$$

$$+ \sum_{i \notin C} \left(\sum_{t=t_i+1}^{t_i+L} \mathbb{E} [Q(\mathbf{w}^{\nu(i)}, y_t) - Q(\mathbf{w}^{\nu(i)}, y^{\nu(i)*})] \right) \quad (38)$$

$$+ \sum_{t=t_i+L+1}^{t_i+L+I} \mathbb{E} [Q(\mathbf{w}^{\nu(i)}, \hat{y}_i) - Q(\mathbf{w}^{\nu(i)}, y^{\nu(i)*})] \quad (39)$$

$$+ \sum_{t=t_i+L+1}^{t_i+L+I} \mathbb{E} [|Q(\mathbf{w}^{\nu(i)}, \hat{y}_i) - Q(\mathbf{w}^{\nu(i)}, y_t)|] \Big). \quad (40)$$

The term in (37) is bounded by $V \times (L+I) \times \max\{b, h\} \log \log I$, because the upper bound for the inventory order-up-to level is $O(\log \log I)$. Similarly, (38) is bounded as

$$\sum_{t=t_i+1}^{t_i+L} \mathbb{E} [Q(\mathbf{w}^{\nu(t)}, y_t) - Q(\mathbf{w}^{\nu(t)}, y^{\nu(t)*})] \leq L \times \max\{b, h\} \log \log I.$$

The term in (39) is bounded as the following. For $i \notin C$, by Assumption 1 (2), there exists a constant $\delta > 0$ such that for any $y \notin \mathcal{Y}^{\nu(i)*}$, $Q(\mathbf{w}^{\nu(i)}, y) - Q(\mathbf{w}^{\nu(i)}, y^{\nu(i)*}) \geq \delta$.

Based on Proposition 3, one has that any $y \in \mathcal{Y}$,

$$\mathbb{P}(|Q(\mathbf{w}^{\nu(i)}, y) - Q(\hat{\mathbf{w}}_i, y)| \leq (3b+2)(\log \log I)^{-\frac{1}{2}}) \geq 1 - \frac{3(\log \log I)^{\frac{1}{4}}}{\theta_2 I^2},$$

therefore, when $2(3b+2)(\log \log I)^{-\frac{1}{2}} < \delta$, one has that

$$\mathbb{P}(Q(\mathbf{w}^{\mathcal{V}(i)}, y^{\mathcal{V}(i)*}) = Q(\mathbf{w}^{\mathcal{V}(i)}, \hat{y}_i)) \geq 1 - \frac{3(\tilde{y}+1)(\log \log I)^{\frac{1}{4}}}{\theta_2 I^2}.$$

Accordingly, define event

$$\mathcal{A}_1 = \{Q(\mathbf{w}^{\mathcal{V}(i)}, y^{\mathcal{V}(i)*}) = Q(\mathbf{w}^{\mathcal{V}(i)}, \hat{y}_i)\},$$

we have

$$\mathbb{P}(\mathcal{A}_1) \geq 1 - \frac{3(\tilde{y}+1)(\log \log I)^{\frac{1}{4}}}{\theta_2 I^2}. \quad (41)$$

Conditioning on $\mathcal{A}_1$, (39) can be bounded as

$$\begin{aligned} & \sum_{t=t_i+L+1}^{t_i+L+I} \mathbb{E} [Q(\mathbf{w}^{\mathcal{V}(i)}, \hat{y}_i) - Q(\mathbf{w}^{\mathcal{V}(i)}, y^{\mathcal{V}(i)*})] \\ &= I \times \left(\mathbb{P}(\mathcal{A}_1) \mathbb{E} [Q(\mathbf{w}^{\mathcal{V}(i)}, \hat{y}_i) - Q(\mathbf{w}^{\mathcal{V}(i)}, y^{\mathcal{V}(i)*}) | \mathcal{A}_1] \right. \\ & \quad \left. + \mathbb{P}(\mathcal{A}_1^c) \mathbb{E} [Q(\mathbf{w}^{\mathcal{V}(i)}, \hat{y}_i) - Q(\mathbf{w}^{\mathcal{V}(i)}, y^{\mathcal{V}(i)*}) | \mathcal{A}_1^c] \right) \\ &\leq I \times \left(0 + \frac{3(\tilde{y}+1)(\log \log I)^{1/4}}{\theta_2 I^2} \times \tilde{y} \max\{b, h\} \right) \\ &\leq \frac{3\tilde{y}(\tilde{y}+1) \max\{b, h\} (\log \log I)^{1/4}}{\theta_2 I}. \end{aligned} \quad (42)$$

The term in (40) represents the regret from missing inventory targets, which occurs when y_t does not equal $\hat{y}_i$. Based on Proposition 4, define the following event

$$\mathcal{A}_2 = \left\{ \omega : \sum_{t=t_i+L+1}^{t_i+L+\log I \log \log I} D_t^{\mathcal{V}(i)} \geq \bar{y} \right\},$$

and one has,

$$\mathbb{P}(\mathcal{A}_2) \geq 1 - \frac{1}{I^2}.$$

On $\mathcal{A}_2$, the cumulative demands during periods $\{t_i + L + 1, \dots, t_i + L + \log I \log \log I\}$ are high enough to consume the initial inventory x_{t_i+L+1} . Therefore, the inventory level will drop to or below $\hat{y}_i$ within the initial $\log I \log \log I$ periods of the exploitation phase, which means that $\hat{y}_i$ can

be reached after $\log I \log \log I$ periods. Therefore, we have

$$\begin{aligned}
& \sum_{t=t_i+L+1}^{t_i+L+I} \mathbb{E} [|Q(\mathbf{w}^{\nu(i)}, \hat{y}_i) - Q(\mathbf{w}^{\nu(i)}, y_t)|] \\
&= \mathbb{P}(\mathcal{A}_2) \mathbb{E} \left[\sum_{t=t_i+L+1}^{t_i+L+I} |Q(\mathbf{w}^{\nu(i)}, \hat{y}_i) - Q(\mathbf{w}^{\nu(i)}, y_t)| \middle| \mathcal{A}_2 \right] \\
&\quad + \mathbb{P}(\mathcal{A}_2^c) \mathbb{E} \left[\sum_{t=t_i+L+1}^{t_i+L+I} |Q(\mathbf{w}^{\nu(i)}, \hat{y}_i) - Q(\mathbf{w}^{\nu(i)}, y_t)| \middle| \mathcal{A}_2^c \right] \\
&\leq \max\{b, h\} \log I \log \log I \times \log \log I + \frac{\max\{b, h\}}{I^2} \times I \times \log \log I \\
&\leq (\max\{b, h\} + 1) \log I (\log \log I)^2.
\end{aligned} \tag{43}$$

This establishes an upper bound for (40).

Putting everything together, we have

$$\begin{aligned}
& \mathcal{R}^\pi(\Lambda, T) \\
&\leq V \times (L + I) \times \max\{b, h\} \log \log I \\
&\quad + \sum_{i \notin C} \left(L \times \max\{b, h\} \log \log I + \frac{3\tilde{y}(\tilde{y} + 1) \max\{b, h\} (\log \log I)^{1/4}}{\theta_2} + (\max\{b, h\} + 1) \log I (\log \log I)^2 \right) \\
&\leq 2 \max\{b, h\} \sqrt{T} \log T (\log \log T)^4,
\end{aligned}$$

and the proof of Theorem 2 is complete. $\square$

Lemma A1. *Let ξ_t be i.i.d. random variables with mean 0 and standard deviation σ . If the moment generating function of ξ_1 around 0 is finite, that is, there exists a constant $\rho > 0$, such that for any $s \in (-\rho, \rho)$ it holds that*

$$\mathbb{E}[e^{s\xi_1}] < +\infty, \tag{44}$$

then one has

$$\mathbb{P} \left(\left| \frac{1}{\log I \log \log I} \sum_{t=1}^{\log I \log \log I} \xi_t \right| \leq 2\sqrt{2}\sigma (\log \log I)^{-1/2} \right) > 1 - \frac{2}{I^2}.$$

Proof. In this proof, let $U = \log I \log \log I$ for simplicity. Under the condition of (44), and following similar lines of proving Lemma A2 in Besbes and Zeevi (2015), Lemma A1 can be proved as the following. For $s \in (-\rho, \rho)$, define

$$\Psi(s) = \log \mathbb{E}[e^{s\xi_1}].$$

For $x > 0$ and $s \in [0, \rho)$, by Markov's inequality one has

$$\mathbb{P}\left(\frac{1}{U} \sum_{t=1}^U \xi_t > x\right) \leq e^{U(\Psi(s) - sx)}.$$

Let $x = 2\sqrt{2}\sigma(\log \log I)^{-1/2}$ and $s^* = x/\sigma^2$. Apply Taylor's expansion to the third order around 0, one has

$$\Psi(s^*) = \frac{1}{2}\sigma^2(s^*)^2 + \frac{1}{6}\Psi'''(s)(s^*)^3,$$

where $s \in [0, s^*]$. Therefore one has

$$\Psi(s^*) - s^*x \leq -\frac{x^2}{2\sigma^2} + c_{67}\frac{x^3}{\sigma^6},$$

where $c_{67} = \max_{s \in [0, s^*]} \Psi'''(s)$. This yields

$$\mathbb{P}\left(\frac{1}{U} \sum_{t=1}^U \xi_t > 2\sqrt{2}\sigma(\log \log I)^{-1/2}\right) \leq e^{U(-\frac{x^2}{2\sigma^2} + c_{67}\frac{x^3}{\sigma^6})} \leq e^{U(-\frac{x^2}{4\sigma^2})} = \frac{1}{I^2}.$$

Similarly, one can prove

$$\mathbb{P}\left(\frac{1}{U} \sum_{t=1}^U \xi_t < -2\sqrt{2}\sigma(\log \log I)^{-1/2}\right) \leq \frac{1}{I^2}.$$

Therefore one has

$$\mathbb{P}\left(\left|\frac{1}{U} \sum_{t=1}^U \xi_t\right| \leq 2\sqrt{2}\sigma(\log \log I)^{-1/2}\right) > 1 - \frac{2}{I^2},$$

and Lemma A1 is thus proved. $\square$